YEAR 12 MATHEMATICS EXTENSION 1

Year 11 Chapter 13 Discrete Probability Distributions

Lesson and Homework Pack

Revision	Probability and data	Year 11 · Ch 12, statistics
Lesson 13A	The language of probability distributions	Past papers: 6 marks · 10 min
Lesson 13B	Expected value	Past papers: none · timed practice
Lesson 13C	Variance and standard deviation	Past papers: 3 marks · 5 min
Lesson 13D	Sampling	Past papers: 1 mark · 5 min
Review	Chapter 13 review exercise	Mixed practice

Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. Then do the **textbook homework** and finish with the **past-paper set** under exam conditions. Answers and full past-paper solutions are in the separate Homework Answers booklet.

CHAPTER REVISION

Probability and data

What this chapter builds on: probability rules (Chapter 12), and the mean, standard deviation and sigma notation (statistics).

SUMMARY · keep this page open while you work

PROBABILITY (CHAPTER 12)

- If E is a subset of a **uniform sample space** S (all outcomes equally likely), then $P(E) = \dfrac{|E|}{|S|}$, where $|E|$ is the number of members of E. An impossible event has probability 0, a certain event has probability 1, and the probabilities of all the outcomes of an experiment add to 1.
- The **complement** $\overline{E}$ is the event 'E does not occur', and $P(\overline{E}) = 1 - P(E)$. Always consider using complementary events, particularly when the word 'not' occurs.
- **Addition rule:** $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. If A and B are **mutually exclusive** (they cannot both occur), then $P(A \text{ or } B) = P(A) + P(B)$.
- **Product rule:** if $A_1, A_2, \ldots, A_n$ are events in successive **independent** stages, then $P(A_1 A_2 \cdots A_n) = P(A_1) \times P(A_2) \times \cdots \times P(A_n)$. Sampling **with replacement** gives independent stages. **Without replacement** the second probability changes, because it is found from the objects that are left. On a **tree diagram**, multiply along a path and add the paths you want.
- **Relative frequency:** if a result occurs f times in n runs of an experiment, its relative frequency is f/n. It estimates the probability.

MEAN, STANDARD DEVIATION AND SIGMA NOTATION (ADVANCED STATISTICS)

- The **mean** of n numbers is found by adding them up and dividing by n: the mean of $0, 1, 2, 3, 4$ is $(0+1+2+3+4)/5 = 2$. In a **frequency table** the value x occurs f times and $n = \sum f$. The **relative frequency** is $f_r = f/n$, and the f_r add to 1. The mean is the mean of the values weighted by their relative frequencies: $\bar{x} = \sum xf / \sum f = \sum x\, f_r$.
- The **standard deviation** σ measures spread. Each **deviation** $x - \bar{x}$ is squared, and the mean of these is the **variance**. It is easier to use the mean of the squares minus the square of the mean:

  $$\sigma^2 = \sum (x - \bar{x})^2 f_r = \sum x^2 f_r - \bar{x}^{\,2} \qquad \sigma = \sqrt{\sigma^2}$$

  On a calculator use the key σ_n.
- **Sigma notation:** $\sum$ is the Greek capital sigma, standing for 'take the sum', so $\sum_{k=1}^{4} k^2 = 1 + 4 + 9 + 16 = 30$. Sums split and constants come out: $\sum (a_k + b_k) = \sum a_k + \sum b_k$ and $\sum c\, a_k = c \sum a_k$.

WARM-UP · answers are in the answers booklet

1 A card is drawn at random from a pack of 52 playing cards (the 12 jacks, queens and kings are the court cards). Find the probability that it is:

a a court card,

b not a court card,

c a heart or a king.

2 A coin is tossed three times. Find the probability that:

a all three tosses are heads,

b there is at least one head,

c there are exactly two heads.

3 Two marbles are drawn in succession from a bag of three red and two blue marbles. Find the probability that both are red if the first marble is:

a replaced before the second is drawn,

b not replaced.

4 A spinner numbered 1 to 4 is spun 50 times. The results are shown in the frequency table.

x	1	2	3	4
f	6	14	18	12

a Find the relative frequency f_r of each value, and check that they add to 1.

b Find the mean $\bar{x}$.

c Find the standard deviation, using $\sum x^2 f_r - \bar{x}^{\,2}$.

5 **a** Write $3 + 6 + 9 + 12$ in sigma notation and evaluate it.

b Given $\sum p(x) = 1$ and $\sum x\,p(x) = 2$, find $\sum (3x + 1)\,p(x)$.

LESSON 13A

The language of probability distributions

A probability distribution lists the possible values of a random experiment together with their probabilities. This lesson sets up the language of the chapter: random experiments and random variables, numeric and categorical distributions, uniform distributions, sample spaces that can be listed (even infinite ones), and the difference between discrete and continuous distributions. Distributions are written as tables, and graphed with the values on the horizontal axis and the probabilities on the vertical axis.

CONCEPT 1 · discrete probability distributions and random variables

WHAT A PROBABILITY DISTRIBUTION IS

Four coins are tossed and the number of heads is recorded. There are five possible outcomes 0, 1, 2, 3 and 4, with different probabilities. Using the methods of the previous chapter, we can calculate all five probabilities and arrange them in a table (see the worked example below).

- The set $\{0, 1, 2, 3, 4\}$ is the **sample space** of the experiment, because each time the experiment is performed, the result will be one, and only one, of these five outcomes. It is not a uniform sample space, however, because these possible outcomes are not equally likely.
- The set of these possible outcomes, together with the corresponding probabilities, is a *probability distribution*. In the context of probability distributions, we usually use the word *values* for the possible outcomes in the sample space.
- This distribution is a *numeric distribution* because all its values are numbers, and it is a *discrete distribution* because it is numeric and its values can be *listed*, which means that they can be written down in some order (here 0, 1, 2, 3, 4).

1 DISCRETE PROBABILITY DISTRIBUTIONS

Suppose that the sample space of an experiment can be *listed*, meaning that it can be written down in some order. Suppose also that the possible outcomes, usually called *values*, are *numeric*.

Then these possible outcomes, together with their probabilities, are called a *discrete probability distribution*.

WHERE THE PROBABILITIES COME FROM

The five probabilities for the four coins come from dividing the results of tossing four coins into 16 equally likely possible outcomes. Many distributions, however, have probabilities that are obtained without any appeal to such a uniform finite sample space. Some probabilities may be obtained by other forms of calculation, and some may be estimates obtained empirically.

RANDOM EXPERIMENTS AND RANDOM VARIABLES

- Throwing four coins, and recording the number of heads, is a *random experiment*, because there is more than one possible outcome. Throwing four coins into an empty bucket, and counting the number of coins in the bucket, is a *deterministic experiment*, because 4 is the only possible outcome.
- Denote by X the number of heads when four coins are thrown. This variable X is called a *random variable*, because it is the result of a random experiment. It can take the values 0, 1, 2, 3 and 4.
- $P(X = 3)$ is read as 'the probability that X takes the value 3'. Similarly we write $P(X > 2)$, $P(X \text{ is odd})$ and $P(X \text{ is prime})$.
- Using x (lower case) for a value of X, the table has a top row x and a bottom row $P(X = x)$. The distribution is **graphed** with the values on the horizontal axis and the probabilities on the vertical axis.

TUTOR DEMO **Two dice, the number of sixes**

Two dice are thrown, and X is the number of sixes showing.
a Write out the probability distribution of X in a table.
b Find $P(X \geq 1)$ and $P(X \text{ is even})$.
c Graph the distribution.

YOUR TURN **Same method**

A die is thrown twice, and X is the number of throws that show a number less than 3.
a Write out the probability distribution of X in a table.
b Find $P(X \leq 1)$ and $P(X \text{ is odd})$.
c Graph the distribution.

WORKED EXAMPLE **From the text: keep for homework**

Four coins are tossed, and X is the number of heads.
a Write out the probability distribution of X in a table.
b Find $P(X = 3)$, $P(X > 2)$, $P(X \text{ is odd})$ and $P(X \text{ is prime})$.
c Graph the distribution.

SOLUTION

a There are $2^4 = 16$ equally likely outcomes, such as HTHH. One has no heads (TTTT), four have one head, six have two heads (HHTT, HTHT, HTTH, THHT, THTH, TTHH), four have three heads, and one has four heads (HHHH). So the table of probabilities is shown on the left. Using x for the number of heads, it can be written as on the right.

number of heads	0	1	2	3	4
probability	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

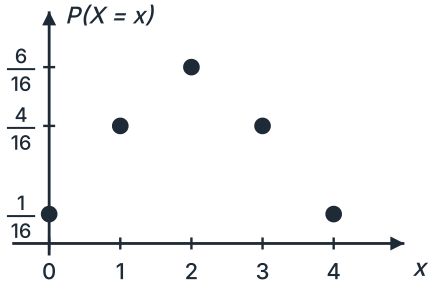
x	0	1	2	3	4
$P(X = x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

b $P(X = 3) = \frac{4}{16}$. This means that the probability that there are three heads is $\frac{4}{16}$. Similarly,

$$P(X > 2) = \frac{4}{16} + \frac{1}{16} = \frac{5}{16},$$
$$P(X \text{ is odd}) = \frac{4}{16} + \frac{4}{16} = \frac{8}{16},$$
$$P(X \text{ is prime}) = \frac{6}{16} + \frac{4}{16} = \frac{10}{16}.$$

(The primes among the values are 2 and 3.)

c The values go on the horizontal axis and the probabilities on the vertical axis.



CONCEPT 2 · categorical data

CATEGORICAL DATA

The values of a probability distribution don't have to be numbers. They may be names, or objects, or any verbal description. The general word for this is 'category', and such data is called *categorical*. For example, if a person is chosen at random from the world's population and the continent he or she lives in is recorded, the values are the six continents (see the worked example below).

2 RANDOM VARIABLE

- A *deterministic experiment* is an experiment with one possible outcome.
- A *random experiment* is an experiment with more than one possible outcome.
- A *random variable*, usually denoted by an upper-case letter such as X, is the result of running a random experiment. It may be *numeric* or *categorical*.

A WORD OF WARNING

The calculations in the next two sections require the values to be numeric, so distributions with categorical values will not greatly concern us. A probability distribution whose values are categorical and can be listed is usually classified as discrete, but in this course, such a distribution is definitely **not** discrete, because it is not numeric.

TUTOR DEMO **Sports houses**

The table shows the number of students in each sports house of a school. A student is chosen at random, and the house is recorded as the value of a random variable X.

House	Red	Blue	Gold	Green	Total
Students	312	288	240	360	1200

a Write out the probability distribution of X in a table.
b Find $P(X = \text{Red or Gold})$.
c Is this distribution numeric or categorical?

YOUR TURN **Same method**

A bag of 50 jelly beans has 18 red, 12 green, 15 yellow and 5 black jelly beans. One jelly bean is taken at random, and its colour is recorded as the value of a random variable X.
a Write out the probability distribution of X in a table.
b Find $P(X = \text{green or yellow})$.
c Is this distribution numeric or categorical?

WORKED EXAMPLE **From the text: keep for homework**

The table to the right was claimed on the web to be the world's population by continent in 2016. A person is chosen at random from the world's population, and the continent he or she lives in is recorded as the value of a random variable X.

a Write out the probability distribution of X.

b Find $P(X = \text{Europe or North America})$.

Continent	Population	Per cent
Asia	4 436 224 000	59.7%
Africa	1 216 130 000	16.4%
Europe	738 849 000	9.9%
North America	579 024 000	7.8%
South America	422 535 000	5.7%
Oceania	39 901 000	0.5%
Total	7 432 663 000	100%

SOLUTION

a This experiment would be very hard indeed to carry out without bias, but we can imagine it. The percentages now become probabilities, which we can arrange in a probability distribution whose values are the six continents (NA and SA stand for North America and South America):

x	Asia	Africa	Europe	NA	SA	Oceania
$P(X = x)$	0.597	0.164	0.099	0.078	0.057	0.005

b $P(X = \text{Europe or North America}) = 0.099 + 0.078 = 0.177$.

The values are continents, not numbers, so this distribution is categorical. In this course it is not called discrete.

CONCEPT 3 · uniform distributions and deterministic experiments

UNIFORM PROBABILITY DISTRIBUTIONS

A discrete or categorical probability distribution is called *uniform* if all its values have the same probability. This means that the values are *equally likely possible outcomes*, and its sample space is a *uniform sample space*. This is what we were dealing with throughout the previous chapter.

3 UNIFORM PROBABILITY DISTRIBUTION

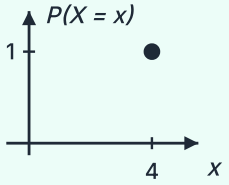
A discrete or categorical probability distribution is called *uniform* if the probabilities of all its values are the same. That is, the values are equally likely possible outcomes, and the sample space is a uniform sample space.

THE PROBABILITY DISTRIBUTION OF A DETERMINISTIC EXPERIMENT

To the right is the trivial probability distribution of the deterministic experiment, 'Throw four coins into an empty bucket and record the number of coins in the bucket.'

x	4
$P(X = x)$	1

The sample space has one possible outcome 4, which is certain to occur, and so has probability 1. The graph is drawn underneath the table. Every deterministic experiment has a similar trivial distribution, and trivially it is a uniform distribution.



TUTOR DEMO **Five numbered cards**

Five cards numbered 2, 4, 6, 8 and 10 are shuffled, and one card is drawn. Write out the probability distribution for the number on the card, and graph the distribution. Is the distribution uniform?

YOUR TURN **Same method**

A tetrahedral die has its four faces numbered 1, 3, 5 and 7. It is thrown, and the number on the face it lands on is recorded. Write out the probability distribution for the number recorded, and graph the distribution. Is the distribution uniform?

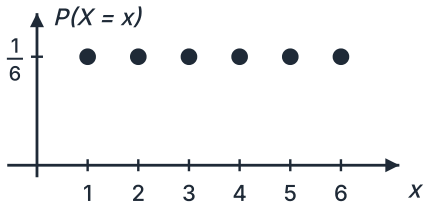
WORKED EXAMPLE 1 **Keep for homework**

A die is thrown. Write out the probability distribution for the number shown on the die, and graph the distribution.

SOLUTION

Each value has probability $\frac{1}{6}$, so the table of the distribution is

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$



CONCEPT 4 · what is a list?

WHAT IS A LIST?

Our definition of discrete probability distribution requires that the values can be listed. Obviously any finite set can be listed, but some infinite sets can also be listed.

- The best example is the set of whole numbers, which can be listed as $0, 1, 2, 3, 4, 5, \ldots$. The list will not terminate, but every whole number will eventually appear once and once only. For this reason, an infinite set that can be listed is called *countably infinite*.
- The integers can be listed, for example as $0, 1, -1, 2, -2, 3, -3, \ldots$.
- But the set of real numbers in an interval such as $0 \leq x \leq 1$ cannot be listed, as the German mathematician Georg Cantor proved in the late 19th century.

4 LISTING THE VALUES OF THE SAMPLE SPACE

- Any finite set can be listed.
- The whole numbers, and the integers, do not form finite sets, but can be listed:
  $0, 1, 2, 3, 4, \ldots$ and $0, 1, -1, 2, -2, 3, -3, \ldots$
- The real numbers in an interval such as $0 \leq x \leq 1$ cannot be listed.

TUTOR DEMO **Throwing until a six**

Ali keeps throwing a die until it shows a six, and X is the number of throws he needs.
a Find $P(X = 1)$, $P(X = 2)$, $P(X = 3)$ and $P(X = 4)$.
b Write out the probability distribution of X in a table, including the general value n.
c Explain why the values of X can be listed, even though there is no largest value.

YOUR TURN **Same method**

A spinner has three equal sectors, one red and two blue. Bea keeps spinning it until it lands on red, and X is the number of spins she needs.
a Find $P(X = 1)$, $P(X = 2)$, $P(X = 3)$ and $P(X = 4)$.
b Write out the probability distribution of X in a table, including the general value n.
c Explain why the values of X can be listed, even though there is no largest value.

WORKED EXAMPLE 2 **Keep for homework**

[An infinite sample space that can be listed]
Wulf is a determined person. He has decided to keep tossing a coin until it shows heads. Adolfa is counting how many times he tosses the coin. What is the probability distribution for this experiment?

SOLUTION

The result of this experiment is how many times Wulf tosses the coin to get a head. If he is very, very unlucky, he may have to toss the coin many, many times.

$$P(\text{Wulf requires 1 toss}) = P(\text{H}) = \tfrac{1}{2}$$
$$P(\text{Wulf requires 2 tosses}) = P(\text{TH}) = \tfrac{1}{4}$$
$$P(\text{Wulf requires 3 tosses}) = P(\text{TTH}) = \tfrac{1}{8}$$
$$P(\text{Wulf requires 4 tosses}) = P(\text{TTTH}) = \tfrac{1}{16},$$

and so on, giving the table:

x	1	2	3	4	5	$\cdots$	n	$\cdots$
$P(X = x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\cdots$	$\frac{1}{2^n}$	$\cdots$

CONCEPT 5 · continuous probability distributions

CONTINUOUS PROBABILITY DISTRIBUTIONS

In a continuous probability distribution, the sample space is typically a closed interval such as $0 \leq x \leq 100$ on the real number line. Examples are the heights or weights of people, and speeds of cars on an expressway.

- In a continuous probability distribution, the probability of any one particular value occurring is zero. The decimal expansion of most real numbers never terminates or repeats.
- So the probabilities involved in a continuous distribution must be recorded as the probabilities that the random variable lies within an interval, for example as $P(55 \leq X \leq 60)$.
- The required machinery for this is *integration*, which is the other process in calculus, and is not covered until Year 12. So continuous probability distributions will not concern us in this chapter.

5 CONTINUOUS PROBABILITY DISTRIBUTIONS

In a continuous probability distribution, the sample space is typically a closed interval such as $0 \leq x \leq 100$ on the real number line.

The sample space is therefore infinite and cannot even be listed.

TUTOR DEMO **Which kind of distribution?**

State whether each probability distribution is numeric or categorical. If it is numeric, state whether it is discrete or continuous.

a The time taken by a randomly chosen student to run 100 metres.

b The number of goals scored in a soccer match.

c The make of a randomly chosen car in a car park.

d The time taken to run 100 metres, recorded by a stopwatch correct to 0.01 seconds.

YOUR TURN **Same skill**

State whether each probability distribution is numeric or categorical. If it is numeric, state whether it is discrete or continuous.

a The mass of a randomly chosen apple.

b The number of children in a randomly chosen family.

c The favourite colour of a randomly chosen student.

d The mass of an apple, recorded by kitchen scales correct to the nearest gram.

WORKED EXAMPLE **From the text: keep for homework**

The police have set up a speed camera to measure the speeds of oncoming cars.

a Is the distribution of the speeds discrete or continuous?

b What is the probability that a car's speed is exactly 56.0123456789 km/h?

SOLUTION

a If we regard the speed as a real number, then we are dealing with a continuous probability distribution. If, however, we take into account that the speed camera only records speeds correct to the nearest 0.01 km/h, then we are dealing with a discrete probability distribution. Such distinctions arise all the time in statistics, because any measurement, no matter how accurate, will only be correct to some number of decimal places.

b In a continuous distribution, the probability of any one particular value is zero. A speed of 56.0123456789 km/h has almost no chance of ever being recorded, even if it could be measured. Instead, we would ask for the probability that the speed lies in an interval, such as $P(55 \leq X \leq 60)$.

TEXTBOOK HOMEWORK · Exercise 13A

FOUNDATION

1 State whether each probability distribution is *numeric* or *categorical*. If it is numeric, state whether it is *discrete* or *continuous*.

a The number showing when a die is thrown.

b The weight of a randomly chosen adult male in Australia.

c Whether it rains or not on a spring day in Sydney.

d The daily rainfall in Sydney on a September day.

e The colour of a ball drawn from a bag containing four red and three green balls.

f The colours of two balls drawn together from a bag containing four red and three green balls.

g The shoe size of a randomly chosen adult female in Australia.

h ATAR results for a particular year.

2 **a** Complete the table for the probability distribution obtained when two coins are thrown one after the other and the successive results are recorded.

outcome	HH	HT	TH	TT
probability				

Look at the probabilities you have obtained. What sort of distribution is this?

b Complete the table for the probability distribution obtained when two coins are thrown and the numbers of heads and tails recorded.

outcome	2 heads	1 head and 1 tail	2 tails
probability			

3 Construct tables for these probability distributions.

a A ball is drawn from a bag containing four red and three green balls, and its colour is noted.

b One hundred tickets are sold in the class lottery. Three friends Jack, Kylie and Lochlan buy six, eight and four tickets respectively. The winner is the person whose ticket is drawn first. Construct the distribution table of the probability that the winner is Jack, Kylie, Lochlan, or Other.

c A letter is chosen at random from the word 'Parramatta'.

d A whole number is chosen at random between 1 and 1000 inclusive, and the number of digits is recorded.

e A whole number between 10 and 19 inclusive is selected at random. A record is made whether it is even, prime, or neither.

4 Each experiment below returns a numerical result. Define a random variable X for each experiment and record its distribution in a table.

a The word lengths in the sentence, 'The ginger cat ran off with the meat.'

b Two coins are thrown and the number of heads recorded.

c A digit is chosen at random from the 12-digit number 1.41421356237.

d Jeff has a collection of marbles with digits on them. He puts marbles numbered 1, 1, 2, 4 into one bag, then puts marbles numbered 2, 3, 3, 5 into a second bag. He randomly selects one of the bags, then randomly selects a marble from that bag and records its number.

5 Amy scoops up coins at random from a pile containing one 10 c coin and two 5 c coins.

a Name the 10 c coin T and the two 5 c coins F1 and F2, and write down as sets the seven possible non-empty scoops.

b Let the random variable X be the value of the money that she picks up. Assuming that she is equally likely to pick up any of these seven non-empty scoops, draw up a probability distribution table for X.

6 Which of these tables are probability distributions? Remember that the probabilities must be all non-negative and add to 1.

a

x	1	2	3	4
$P(X=x)$	0.1	0.6	0.2	0.1

b

x	1	2	3	4
$P(X=x)$	0.5	0.3	-0.2	0.4

c

x	1	2	3	4
$P(X=x)$	0.25	0.25	0.25	0.25

d

x	1	2	3	4
$P(X=x)$	30%	20%	40%	10%

e

x	1	2	3	4
$P(X=x)$	0.7	0.2	0.4	0.2

f

x	1	2	3	4
$P(X=x)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{4}$

7 A discrete probability distribution is tabulated below.

x	0	1	2	3	4	5
$P(X=x)$	0.1	0.2	0.3	0.25	0.05	0.1

Find:

a $P(X=1)$

b $P(2 \leq X \leq 4)$

c $P(1 \leq X < 4)$

d $P(X=6)$

e $P(X \leq 2)$

f $P(X<4)$

g $P(X \geq 1)$

h $P(X>1)$

i $P(X \text{ is even})$

DEVELOPMENT

8 These questions all require probability tree diagrams with identical stages.

a A pack of cards has 52 cards, of which 12 are court cards (jack, queen or king). A card is drawn at random, and it is recorded whether or not it is a court card. The card is replaced, then a second card is drawn and again it is recorded whether or not it is a court card.
(i) Draw up a probability tree diagram with two identical stages.
(ii) Let X be the number of court cards drawn. Use the results of part (i) to draw up a probability distribution table for X.

b Three dice are thrown, and the number of even numbers is recorded.
(i) Draw up a probability tree diagram with three identical stages, each stage being, 'Throw a die and record whether the result is even or odd'.
(ii) Let X be the number of even numbers when the three dice are thrown. Use the results of part (i) to draw up a probability distribution table for X.

c A class has 10 girls and 15 boys. Three times the teacher chooses a student at random to answer a question, not caring whether he has called that name before. Use a similar method to draw up a probability distribution table for the number of girls called.

d The Spring Hill Zoo has 20 friendly wallabies. Six are from Snake Gully, five are from Dingo Ridge, and nine are from Acacia Flat. On three days last week the zookeepers selected a wallaby at random and took it to breakfast with them, then returned it to the enclosure. Draw up a probability distribution table for the number X of Snake Gully wallabies taken to breakfast.

9 Find the unknown constant a in these probability distributions. Use the facts that $0 \leq P(X = x) \leq 1$ for each value x, and that the sum of the probabilities is 1.

a

x	1	2	3	4	5
$P(X = x)$	$4a$	$2a$	$9a$	$3a$	$7a$

b

x	1	2	3	4	5
$P(X = x)$	$3a$	a	$4a$	a	$5a$

c

x	-2	-1	0	1	2
$P(X = x)$	a	$3a$	$5a$	$7a$	$11a$

d

x	10	20	30	40	50
$P(X = x)$	$1 - 3a$	a	$1 - 9a$	$1 - 10a$	a

e

x	1	2	3	4	5
$P(X = x)$	$0.2a$	$0.1a$	$0.5a$	$0.1a$	$0.1a$

10 These questions all involve probability tree diagrams in which the stages are different because the sampling has been done 'without replacement'.

a A bag contains six marbles numbered 1, 2, 3, 4, 5, 6. A marble is drawn, and it is recorded whether the number is odd or even. Without replacing the marble, a second marble is drawn, and it is recorded again whether the number is odd or even.
(i) Draw up a probability tree diagram for the four outcomes.
(ii) Draw up a probability distribution table for the number X of even numbers chosen.

b Two students are chosen from a group of four boys and two girls. Use a similar method to draw up a probability distribution table for the number X of girls chosen.

c Five tiles marked E, E, E, R, T are turned upside down, and two are selected at random one after the other.
(i) Draw a probability tree diagram for the seven possible selections of two tiles.
(ii) Draw up a probability distribution table for the number X of Es selected.

11 A small pack of cards consists of three 4s, two 2s and one 5 (the suit is not important for this experiment). A card is selected at random and the value recorded. It is then returned to the pack. This is repeated. Complete the probabilities for the outcomes of the categorical random variable X in the following table.

x	22	44	55	24 or 42	25 or 52	45 or 54
$P(X = x)$						

[This is an example of a *multinomial distribution*, a multi-stage experiment with identical stages, each with multiple outcomes. Here there are two stages, because we select a card twice, and the stages are identical because the card is replaced.]

12 Construct tables for the following distributions. In each case the outcome is categorical, namely a pair of colours or a pair of suits.

a A ball is drawn from a bag containing four red and three green balls, and the colour is recorded. The ball is returned, a second ball is drawn, and the colour also recorded.

b A ball is drawn from a bag containing four red and three green balls, and the colour is recorded. A second ball is drawn without replacement and the colour also recorded.

c A pair of cards are drawn from a pack and the suits are noted (but not their order).

13 [Simulation experiment]

Cut five identical pieces of paper or cardboard. Label three of them *Red* and the other two *Green*. Place the pieces in a hat or bag.

In this experiment, two pieces of paper are drawn out, the number of greens is recorded, and the pieces are then returned. Let X be the random variable for the number of greens.

a Copy the table below for recording the results of your experiment.

x	0	1	2
Tally			

b Repeat the experiment 40 times, recording each outcome in the tally column.

c Add two rows to your table, as below. Complete the table to calculate the experimental probabilities of each outcome.

x	0	1	2
Tally			
Frequency f			
Relative frequency $f_r = \frac{f}{40}$			

d Compare your results with other members of the class. Does this suggest that the results are accurate? Can you explain any differences in the results obtained?

e Calculate the theoretical probabilities for this experiment and add this as a further row to your table. How do your results compare?

f Comment on any aspects of your experimental design that assist the accuracy of the results. Is there any way you could have improved the reliability and randomness of your experiment?

ENRICHMENT

14 A bag has six marbles marked 1, 2, 3, 4, 5, 6. Three marbles are drawn in succession, without being replaced, and the number of even-numbered marbles is recorded. Let X be the number of marbles with even numbers chosen. Draw up a probability distribution table for X.

15 Find the unknown constant a in these probability distributions.

a

x	1	2	3	4	5
$P(X=x)$	$a(a+1)$	$3a^2$	$1-3a$	$1-4a$	a

b

x	1	2	3	4
$P(X=x)$	$\frac{1}{6}(a+1)$	$\frac{1}{4}a^2$	$\frac{1}{8}(5-3a)$	$\frac{1}{6}(3-2a)$

16 **a** Kylie has a hand of four cards: the 7 of hearts, 7 of diamonds, 6 of clubs and 8 of spades. She takes three of the cards at random, places them on the table, and adds the cards' values. Construct a table showing the probability distribution of the sum.

b Repeat this experiment if Kylie has a hand of five cards, including three 7s, a 6 and an 8, and still chooses three cards at random.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 10 minutes **6 marks** Calculator allowed · show working · hints are below each question if you get stuck

School trial papers rarely test this lesson on its own. These items are parts of longer questions: only the parts that use this lesson's ideas are included.

PP 1 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(c)(i)** 1 mark

Two fair six-sided dice are rolled together and the sum of the numbers on their upper faces is recorded. The result is recorded as either a 7 or not a 7.

(i) If p is the probability of a sum of 7, find the value of p. 1

Hint: how many of the 36 equally likely outcomes give a sum of 7?

PP 2 **Baulkham Hills High School, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q12(a)(i)(iv)** 2 marks

The following table is the distribution of a random variable X where $X \sim \text{Bin}(4, p)$.

x	0	1	2	3	4
$P(X = x)$	0.2401	a	b	0.0756	0.0081

(i) Use the known entries in the probability distribution table to calculate the value of p. 1

(iv) Find $P(X \geq 3)$. 1

Hint: $X \sim \text{Bin}(4, p)$ means that X is the number of successes in 4 independent trials, each with probability p of success. For (i), which entry of the table is easiest to link to p? For (iv), the table is all you need.

PP 3 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)(ii)** 1 mark

A test consists of five multiple choice questions. Each question has four alternative answers. For each question only one of the alternative answers is correct. Jerry randomly selects an answer to each of the five questions.

(ii) What is the probability that Jerry selects at least one incorrect answer? 1

Hint: think about the complementary event.

PP 4 **Northern Beaches Secondary College Manly Selective Campus, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q11(a)(iii)** 2 marks

On average an archer scores a bullseye once in every three shots at the target. A competitor is allowed eight shots at the target.

(iii) How many shots should the archer use so as to be 90% certain that at least one bullseye is hit? 2

Hint: think about the complementary event to 'at least one bullseye'.

LESSON 13B

Expected value

The expected value $\mathrm{E}(X)$ of a discrete probability distribution is the weighted mean of its values, each value weighted by its probability: $\mathrm{E}(X) = \sum xp(x)$. It is a measure of central tendency, and is also called the mean μ of the distribution. This lesson introduces the function notation $p(x)$, sets out the calculation of $\mathrm{E}(X)$ in a table with an extra row (or column), and finds the expected value of uniform distributions. The expected value need not be one of the values of the distribution.

CONCEPT 1 · the notation p(x) and the centre of a distribution

THE NOTATION $p(x)$

The first experiment in Lesson 13A was, 'Toss four coins and count the number of heads':

x	0	1	2	3	4
$P(X = x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

The notation $P(X = x)$ will be too unwieldy for what follows, so we will write it instead using standard function notation as $p(x)$. For each value x in the distribution, $p(x)$ means 'the probability that $X = x$'.

$p(x) = P(X = x)$, for all values x in the distribution.

With this more concise notation, the table becomes:

x	0	1	2	3	4
$p(x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

WHERE IS THE CENTRE OF THE DISTRIBUTION?

Someone looking at this table may want to say, '*I would expect to get the result two heads.*' There are two good ways to explain this, and they are completely different (see the worked example below). One is based on running the experiment a large number of times and averaging things out. The other is based solely on the symmetry of the distribution's graph.

This section introduces a clearer and more general way of dealing with this 'expected value', which is better described as the 'mean' of the distribution. From now on, values must be numeric and not categorical.

TUTOR DEMO **The sum of two dice**

Two dice are thrown, and X is the sum of the two numbers showing.
a Write out the probability distribution of X in a table, using the notation $p(x)$.
b Graph the distribution.
c Use the symmetry of the graph to find the centre of the distribution.

YOUR TURN **Same method**

Two tetrahedral dice, each with its four faces numbered 1, 2, 3 and 4, are thrown, and X is the sum of the two numbers on the faces they land on.
a Write out the probability distribution of X in a table, using the notation $p(x)$.
b Graph the distribution.
c Use the symmetry of the graph to find the centre of the distribution.

WORKED EXAMPLE **From the text: keep for homework**

Four coins are tossed, and X is the number of heads. Where is the centre of the distribution?

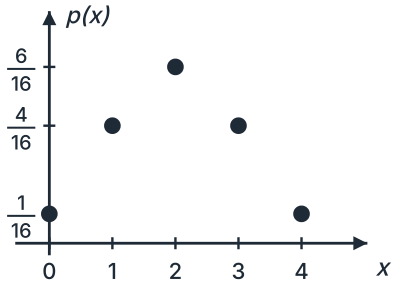
SOLUTION

Someone looking at the table of $p(x)$ above may want to say,

> *'I would expect to get the result two heads.'*

When asked, she may try to explain what she means by saying,

> *'When we do the experiment a large number of times, the average number of heads will be about 2, and the more times we do it, the closer the average will be to two'.*



This is a good explanation, because it is based on the interpretation of probability as running the experiment a large number of times and averaging things out.

She may also try to explain it by symmetry:

> *'If I plot the five probabilities, the pattern is symmetric about the value 2'.*

This is also a good explanation, but is completely different. It ignores the idea of running trials, and is based solely on the symmetry of the distribution's graph. It observes that each smaller number of heads on the left of $x = 2$ is balanced out by a larger number of heads on the right of $x = 2$ with the same probability, so $x = 2$ can be regarded as the centre of the distribution.

CONCEPT 2 · expected value, the weighted mean

EXPECTED VALUE

Everyone is familiar with the idea of the mean or average of a set of n numbers: add them up and divide by n,

$$\text{mean} = \frac{0+1+2+3+4}{5} = \frac{10}{5} = 2.$$

Alternatively, multiply each number by $\frac{1}{n}$ and add the results:

$$\text{mean} = 0 \times \tfrac{1}{5} + 1 \times \tfrac{1}{5} + 2 \times \tfrac{1}{5} + 3 \times \tfrac{1}{5} + 4 \times \tfrac{1}{5} = \tfrac{10}{5} = 2.$$

THE WEIGHTED MEAN

But to calculate the mean of a discrete probability distribution, we need a *weighted mean* of the values, giving more *weight* to more likely values, which will occur more often, and less weight to the less likely values, which will occur less often. So instead of multiplying each value by $\frac{1}{n}$, we *weight* each value by its probability,

$$\begin{aligned}\textbf{weighted mean} &= 0 \times \tfrac{1}{16} + 1 \times \tfrac{4}{16} + 2 \times \tfrac{6}{16} + 3 \times \tfrac{4}{16} + 4 \times \tfrac{1}{16} \\ &= 0 + \tfrac{4}{16} + \tfrac{12}{16} + \tfrac{12}{16} + \tfrac{4}{16} \\ &= \tfrac{32}{16} \\ &= 2, \ \textbf{which is called the } \textit{expected value } \mathrm{E}(X).\end{aligned}$$

This process is called *weighting the values by their probabilities*, and is the way the expected value $\mathrm{E}(X) = 2$ of this distribution is calculated. Here the result 2 happens to be the same as the ordinary mean of the five values, but this is only because of the mirror symmetry of the probabilities about $x = 2$.

THE FORMULA FOR EXPECTED VALUE

The usual notation for this procedure will be completely unfamiliar. In any distribution, the weighting of a value x by its probability $p(x)$ means that we take the product of the value x and its probability $p(x)$, written as the product

$$xp(x).$$

Now we need to add up all these products, for all the values in the distribution. The symbol $\sum$ is the mathematical symbol for 'take the sum'. It is the Greek upper-case sigma corresponding to the Latin letter 'S' and standing for 'sum'. Thus the formula for the expected value $\mathrm{E}(X)$ is written as

$$\mathrm{E}(X) = \sum xp(x), \text{ summing over all values of the distribution.}$$

The phrase, 'summing over all values of the distribution' means that we take all the products $xp(x)$, for all the values of the distribution, and add them up.

SETTING OUT THE CALCULATION OF EXPECTED VALUE

The actual calculation is best set out not as shown above, nor by substituting into the formula, but in tabular form, extending the table of values and probabilities of the distribution with an extra row and an extra column:

x	0	1	2	3	4	Sum
$p(x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$	1
$xp(x)$	0	$\frac{4}{16}$	$\frac{12}{16}$	$\frac{12}{16}$	$\frac{4}{16}$	2

as we obtained before.

The probabilities must add to 1. Summing the middle row is only a check.

Question 3 of Exercise 13B is an investigation that involves throwing four coins a large number of times and seeing how close the average value is to the expected value 2 calculated above.

AN ALTERNATIVE SETTING OUT USING COLUMNS RATHER THAN ROWS

Many people prefer to use columns rather than rows for these calculations. There is a very good reason for this: it is easier to add up a column of numbers than to add up a row of numbers.

If you use columns rather than rows, then your table will be *transposed*, meaning that the rows become columns and the columns become rows.

This setting out is shown to the right.

x	$p(x)$	$xp(x)$
0	$\frac{1}{16}$	0
1	$\frac{4}{16}$	$\frac{4}{16}$
2	$\frac{6}{16}$	$\frac{12}{16}$
3	$\frac{4}{16}$	$\frac{12}{16}$
4	$\frac{1}{16}$	$\frac{4}{16}$
Sum	1	2

EXPECTATION IS A MEASURE OF CENTRAL TENDENCY

As mentioned earlier, the expected value $\mathrm{E}(X)$ of a distribution is also called the *mean* of the distribution, because it is obtained as the weighted mean of the values. It is a *measure of central tendency*, meaning that it is a measure of where the middle of the distribution lies.

For this reason, the expected value $\mathrm{E}(X)$ is often assigned the pronumeral μ, a Greek lower-case letter corresponding to Latin 'm', and standing for 'mean'.

6 EXPECTED VALUE OF A DISCRETE PROBABILITY DISTRIBUTION

The *expected value* of a discrete probability distribution with numeric data is the weighted mean of the values x weighted by the probabilities $p(x)$,

$\mathrm{E}(X) = \sum xp(x)$, summing over all values of the distribution.

- The symbol $\sum$ is the Greek upper-case sigma, and stands for 'sum'.
- Each term in the sum is the product $xp(x)$ of a value and its probability.
- Take all these products $xp(x)$, for all the values of the distribution, and add them up.
- The expected value is also called the *mean* of the distribution. It is usually assigned the pronumeral μ. The Greek letter 'mu' stands for 'mean'.
- The expected value is a *measure of central tendency*.

UNSYMMETRIC DISTRIBUTIONS

The example above is symmetric, which is why the mean of the values turned out to be the same as the expected value. Worked example 3 shows how well the idea of expected value is able to pick out the central tendency in quite unsymmetric data.

TUTOR DEMO **The pickpocket**

Sixteen friends meet at a café. One has $300 cash, three have $120 cash, two have $90 cash, four have $40 cash, and two have $20 cash. The other four have no cash. A pickpocket chooses one of the friends at random and takes all of his or her cash. Graph the distribution, and find the pickpocket's expected gain.

YOUR TURN **Same method, using columns**

Ten friends are on a bus. Two have $\$50$ cash, three have $\$30$ cash, and one has $\$20$ cash. The other four have no cash. A thief chooses one of the friends at random and takes all of his or her cash. Set out the calculation in columns, graph the distribution, and find the thief's expected gain.

WORKED EXAMPLE 3 **Keep for homework**

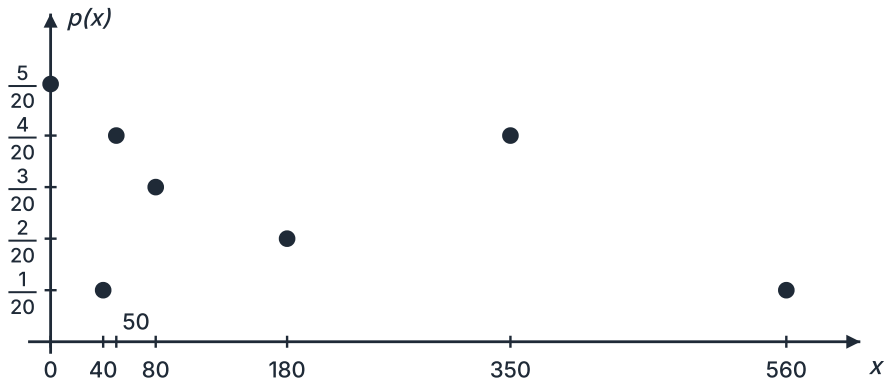
Twenty friends go out to dinner at a Vietnamese restaurant. One has $\$560$ cash, four have $\$350$ cash, two have $\$180$ cash, three have $\$80$ cash, four have $\$50$ cash, and one has $\$40$ cash. Five have no cash, and are expecting to borrow from the others. An armed robber bursts in and seizes one of the friends at random. She threatens him, grabs all his cash, and runs away. Graph the distribution, and find her expected criminal gain.

SOLUTION

Let $\$x$ be the amount the criminal seizes. The probability distribution with the added row and column is

x	0	40	50	80	180	350	560	Sum
$p(x)$	$\frac{5}{20}$	$\frac{1}{20}$	$\frac{4}{20}$	$\frac{3}{20}$	$\frac{2}{20}$	$\frac{4}{20}$	$\frac{1}{20}$	1
$xp(x)$	0	2	10	12	18	70	28	140

Hence her expected criminal gain is $\$140$.



CONCEPT 3 · expected value of a uniform distribution

EXPECTED VALUE OF A UNIFORM DISTRIBUTION

In a uniform distribution, all the values have the same probability. We could add the values and then divide by the number of values, but it is better to continue with the same setting out. There are two things to notice here:

- The sample space of a uniform distribution consists of equally likely possible outcomes. These were the sorts of sample spaces that most of the calculations in Chapter 12 were based on.
- In a uniform distribution with n values, each value has probability $\frac{1}{n}$. Thus to weight each value by its probability, we multiply it by $\frac{1}{n}$, so that calculating the expected value is the same as calculating the mean of the values.

TUTOR DEMO **An octahedral die**

An octahedral die has its eight faces numbered 1 to 8. It is thrown, and the number on the face it lands on is recorded. Graph the distribution and find its expected value. Is the expected value one of the numbers on the die?

YOUR TURN **Same method**

A dodecahedral die has its twelve faces numbered 1 to 12. It is thrown, and the number on the face it lands on is recorded. Graph the distribution and find its expected value. Is the expected value one of the numbers on the die?

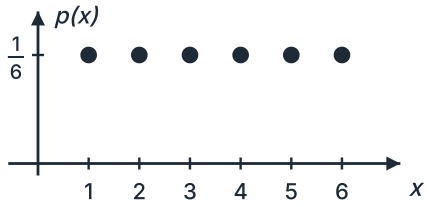
WORKED EXAMPLE 4 **Keep for homework**

A die is thrown. Graph the distribution and find its expected value.

SOLUTION

Each value has probability $\frac{1}{6}$, so the table is

x	1	2	3	4	5	6	Sum
$p(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	1
$xp(x)$	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	$\frac{6}{6}$	$\frac{21}{6}$



Hence the expected value on the die is $\mu = 3\frac{1}{2}$.

Do you find this answer just a little unsettling? How can we 'expect' an answer that we will never see on the die?

The same thing happened in the restaurant robbery in Worked example 3: the expected value was $\$140$, but no one amongst the friends had $\$140$ cash. There is actually no problem here, because the expected value is just a fancy name (and perhaps a slightly misleading name) for the mean of the distribution, and we know from countless examples that the mean of a set of numbers is usually not one of the numbers.

TEXTBOOK HOMEWORK · Exercise 13B

FOUNDATION

1 Complete each table and find the expected value $\mathbf{E}(X)$ for the distribution, where the function $p(x)$ is defined as usual by $p(x) = P(X = x)$, for all values x.

a

x	0	1	2	3	Sum
$p(x)$	0.4	0.1	0.2	0.3	
$xp(x)$					

b

x	2	4	6	8	Sum
$p(x)$	0.1	0.4	0.4	0.1	
$xp(x)$					

c

x	-50	-20	0	30	100	Sum
$p(x)$	0.1	0.35	0.4	0.1	0.05	
$xp(x)$						

2 A simple gambling game involves the throw of a die. Players are charged 40 cents if the die shows 1, 2 or 3, they are charged nothing for 4, and they receive 30 cents or 60 cents respectively for 5 or 6. Let the random variable X be the payout to the player.

a Copy and complete the table to the right written this time in columns instead of rows.

b Calculate the expected value by summing the third column.

c What does this expected value represent?

d How much profit would the casino expect to make on 100 games?

x	$p(x)$	$xp(x)$
-40		
0		
30		
60		
Sum	1	

3 Four coins are tossed and the number of heads recorded. In the theory for this section, we constructed the table reproduced below. The expected value was calculated to be 2.

x	0	1	2	3	4	Sum
$p(x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$	1
$xp(x)$	0	$\frac{4}{16}$	$\frac{12}{16}$	$\frac{12}{16}$	$\frac{4}{16}$	2

In this question we shall simulate the experiment and see if we have experimental agreement with these results.

a Toss four coins and record the number of heads, using a tally row and filling in a copy of the table below. Repeat this experiment 32 times.

x	0	1	2	3	4	Sum
Tally						—
Frequency f						
Relative frequency $f_r = \frac{f}{32}$						

b Check whether your frequencies agree reasonably with the results 2, 8, 12, 8, 2 that the theoretical distribution would predict. Check also whether the relative frequencies agree reasonably with the corresponding probabilities 0.0625, 0.25, 0.375, 0.25, 0.0625. If your results do not agree closely, you might like to repeat the experiment a further 16 or 32 times (and divide by 48 or 64 to calculate the relative frequencies.)

c Using your calculator, or by hand using the table below, calculate the mean for this data using the values x and their frequencies.

x	0	1	2	3	4	Sum
Frequency f						
$x \times f$						

$$\text{Mean} = \frac{\text{sum of } xf}{\text{sum of frequencies}}$$
$$= \ldots$$

d Does your answer for the mean approximate the theoretical expected value $\mathrm{E}(X) = 2$?

4 [An alternative notation]
Sometimes the values of a discrete probability experiment are indexed $x_1, x_2, x_3, \ldots$, with corresponding probabilities $p_1, p_2, p_3, \ldots$. These values and probabilities can be abbreviated to x_i and p_i, where i is called the *index* variable because it *indexes* the values and probabilities. The calculations then proceed as before.
Find the expected value for these distributions.

a

x_i	2	4	6	8	10	Sum
p_i	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	
$x_i p_i$						

b

x_i	-3	1	2	5	6	Sum
p_i	0.1	0.3	0.2	0.3	0.1	
$x_i p_i$						

DEVELOPMENT

5 Fiona's mother tells her that she may buy a pencil, eraser or pen from the stationery shop. A pencil costs $\$1.50$, an eraser costs $\$2.10$ and a pen costs $\$2.40$. There are five boxes of pencils, four boxes of erasers and three boxes of pens.

a If she chooses a box at random and takes one item from the box, what is the expected cost?

b If Fiona and her 99 friends each choose at random, what would be the expected total cost?

6 In this question we investigate what happens to the expected value if we transform the random variable, such as by doubling all the values, or increasing them all by 1. Because we are dealing with more than one random variable, we have retained the notation $P(X = x)$, $P(Y = y)$, and so on from Lesson 13A.

A random variable X records the outcomes of a spinner with sectors labelled 1, 2, 3, 4. The spinner is biased because it has been weighted.

x	1	2	3	4
$P(X = x)$	0.1	0.1	0.5	0.3
$x \times P(X = x)$				

a Copy and complete the table to calculate $\mathbf{E}(X)$.

y	2	4	6	8
$P(Y = y)$	0.1	0.1	0.5	0.3
$y \times P(Y = y)$				

b A second random variable defined by $Y = 2X$ records twice the outcome of the weighted spinner.

(i) Calculate $\mathbf{E}(Y)$ from this table.

(ii) Does your result agree with the result $\mathbf{E}(aX) = a\mathbf{E}(X)$?

z	2	3	4	5
$P(Z = z)$	0.1	0.1	0.5	0.3
$z \times P(Z = z)$				

c A third random variable defined by $Z = X + 1$ records the outcome of the weighted spinner plus one.

(i) Calculate $\mathbf{E}(Z)$ from this table.

(ii) Does your result agree with the result $\mathbf{E}(X + b) = \mathbf{E}(X) + b$?

The general result that these examples illustrate is

$$\mathbf{E}(aX + b) = a\mathbf{E}(X) + b, \quad \text{for all constants } a \text{ and } b.$$

7 A random variable X is known to have the property that $\mathbf{E}(X) = 5$. Use the formula $\mathbf{E}(aX + b) = a\mathbf{E}(X) + b$ to calculate:

a $\mathbf{E}(3X)$

b $\mathbf{E}(X + 5)$

c $\mathbf{E}\left(\frac{1}{2}X\right)$

d $\mathbf{E}(X - 2)$

e $\mathbf{E}(10 - 2X)$

f $\mathbf{E}(4X - 2)$

8 A coin is tossed three times and the number of heads is recorded. Construct a table showing the probability distribution, and calculate the expected value.
(This is an example of a *binomial distribution*, a multi-stage experiment with identical stages, where at each stage there are two possible outcomes. Here there are three stages, because we toss the coin three times and there are two possible outcomes, heads or tails, for each throw.)

9 Two cards are selected at random from a standard pack, and the number of hearts is recorded. Note that this is equivalent to selecting two cards without replacement. Construct a table showing the probability distribution and calculate the expected value.
(This is an example of a *hypergeometric distribution*, a multi-stage experiment involving two possible outcomes, where at each stage the object is selected without replacement. Here there are two stages because we select two cards.)

10 [Simulation experiment]
Two dice are thrown. Let X be the difference between the two resulting numbers, so that the sample space consists of the integers 0, 1, 2, 3, 4, 5.

a Conduct an experiment to determine the experimental probability of the six outcomes. You should conduct the experiment 36 times and record your results in a copy of the table below.

x	0	1	2	3	4	5	Sum
Tally							—
Frequency f							
Relative frequency $f_r = \frac{f}{36}$							
$x \times f_r$							

b Use the relative frequencies as estimates for the probabilities of each outcome. Calculate the experimental expected value (the symbol for this experimental expected value is $\bar{x}$ to distinguish it from the theoretical expected value μ).

c Compare your results with other members of the class.

d (i) Calculate the theoretical probabilities, using the normal 6×6 array of dots.
(ii) Also calculate the theoretical expected value by copying and completing the table below.

x	0	1	2	3	4	5	Sum
$p(x)$							
$xp(x)$							

e How do your results compare?

f If you think your results are inaccurate, consider any design faults in your experiment.

g Do your results agree more closely with the theoretical probabilities if you increase the number of trials?

ENRICHMENT

11 The backers of the game in Question 2 have weighted the die so that it is 50% more likely to turn up each of the results 1, 2 or 3 as to turn up 4, 5 or 6. (This is illegal.)

a What are the probabilities now of each outcome?

b Find the expected value now.

12 A company is designing a new gambling slot machine for a casino. A player pulls a lever and the machine randomly rolls three outcomes from Orange, Strawberry and Apple, for example AOS or SSO. The probabilities of the various fruits turning up are in the ratio $1:2:3$, so that Apple is three times as likely to occur in a given position as Orange. The machine pays out in the ratio $11:2:1$ if it turns up three Oranges, three Strawberries or three Apples, respectively. No other combination pays the player. A player initiates a roll by feeding a $\$1$ coin into the machine.

a Find the probabilities that Orange, Strawberry or Apple turn up in a given position.

b Suppose the triple AAA pays $\$k$. Construct a probability distribution table showing the outcomes OOO, SSS, AAA and Other for the random variable X representing the payout.

c Determine how much a player would get for the highest paying outcome OOO, if the machine is designed so that it will just break even (this is unlikely).

13 [Expected value when the sample space is infinite, but can be listed]
In Worked example 2 of Lesson 13A, Wulf tosses a coin repeatedly until it shows a head. The probability table for the number X of tosses is infinite:

x	1	2	3	4	5	$\cdots$	n	$\cdots$
$p(x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\cdots$	$\frac{1}{2^n}$	$\cdots$

Thus to calculate the expected value we need to calculate the infinite sum:

$$\mu = 1 \times \tfrac{1}{2} + 2 \times \tfrac{1}{4} + 3 \times \tfrac{1}{8} + 4 \times \tfrac{1}{16} + 5 \times \tfrac{1}{32} + 6 \times \tfrac{1}{64} + \cdots$$

a Write down the sum for 2μ and by carefully subtracting like fractions, show that

$$2\mu - \mu = 1 + \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{16} + \tfrac{1}{32} + \cdots$$

and hence write down a second infinite sum for μ.

b Repeat the process of doubling and subtracting with this new infinite sum for μ, and hence calculate μ.

c Explain what this expected value means practically and design an experiment to test your theoretical result.

Note: The operations on infinite series used here are valid for these particular series because they are convergent (this concept is explained in Year 12, and even then the proofs are difficult). The operations are certainly not true for all infinite series.

14 [St Petersburg Paradox]

Patrons at a casino may play a game involving a single coin, which is tossed until a head turns up. A jackpot is set aside. It initially contains $2, but its value is doubled on each toss of the coin turning up a tail. The winner receives the contents of the pot when the first head is tossed. Thus the player would win $2 if the initial toss is a head, $4 if the second toss is the first head, $8 if the third toss is the first head, and so on.

A manager at the casino suggests that a player should be charged $40 to play the game. Calculate the player's expected return, and comment on the manager's advice.

PAST-PAPER HOMEWORK · timed practice

The school trial papers used for this pack (2020–2024) have no question on expected value that can be done without the Year 12 binomial results. For timed practice, do Questions 8, 9, 11 (after Question 2) and 12 above in 25 minutes.

LESSON 13C

Variance and standard deviation

The variance $\mathrm{Var}(X)$ of a discrete probability distribution measures how spread out its values are from the mean μ. It is the weighted mean of the squared deviations $(x-\mu)^2$, so $\mathrm{Var}(X) = \mathrm{E}\big((X-\mu)^2\big)$. The alternative formula $\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2$ is usually easier to calculate. The standard deviation $\sigma = \sqrt{\mathrm{Var}(X)}$ has the same units as the values. This lesson sets out both formulas in tables with extra rows, proves the alternative formula, and uses the standard deviation to compare scores.

CONCEPT 1 · the variance, first formula

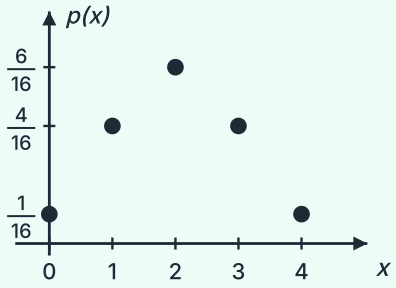
MEASURES OF SPREAD

After discussing $\mathrm{E}(X)$, which is a measure of the central tendency of a discrete probability distribution, we now turn to variance and standard deviation, which are two measures of spread. Variance is easier to define and calculate, so we will work first with the variance. But standard deviation, which is just the square root of the variance, is important because it has the same units as the values of the distribution, and increases in proportion when all the values are increased by some factor.

THE VARIANCE: FIRST FORMULA

Return again to the number of heads when four coins are tossed:

x	0	1	2	3	4
$p(x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$



In Lesson 13B, the expected value of this distribution, its mean, was found to be $\mu = 2$. We now want a measure of how spread out the values are from the mean. The usual measure of this is the *variance* $\mathrm{Var}(X)$.

DEVIATIONS FROM THE MEAN

To find the variance, take the difference $x - \mu$ of each value from the mean μ. This difference is called the *deviation* of the value x from the mean μ. Then we square the deviation to give $(x - \mu)^2$. This squared deviation is a good measure of how far x is from the mean for two reasons:

- The square $(x - \mu)^2$ is always a positive number or zero, whether the value x is on the left or the right of μ, because of the squaring.
- The square $(x - \mu)^2$ gets larger as x moves away from the mean, and the square means it gets larger very quickly. For example, doubling the distance of x from the mean has four times the effect on the square.

Then, as with expected value, take the weighted mean of these squared deviations $(x - \mu)^2$, weighted as before by the probabilities. Using sigma notation,

$$\text{Var}(X) = \sum (x - \mu)^2 p(x), \quad \text{summed over the distribution.}$$

SETTING OUT THE CALCULATION OF VARIANCE

As with expected value, the calculation is best done by adding more rows to the probability distribution table. Here are the calculations for the number of heads on four coins. The table calculates the expected value and variance together:

x	0	1	2	3	4	Sum	
$p(x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$	1	(a check)
$xp(x)$	0	$\frac{4}{16}$	$\frac{12}{16}$	$\frac{12}{16}$	$\frac{4}{16}$	2	(the mean μ)
$(x - \mu)^2$	4	1	0	1	4	—	
$(x - \mu)^2 p(x)$	$\frac{4}{16}$	$\frac{4}{16}$	0	$\frac{4}{16}$	$\frac{4}{16}$	1	(the variance)

Thus the mean is 2 and the variance is 1.

USING COLUMNS INSTEAD OF ROWS

To the right is the alternative setting out that uses columns instead of rows.

Take your pick: use the layout that you find more convenient.

x	$p(x)$	$xp(x)$	$(x - \mu)^2$	$(x - \mu)^2 p(x)$
0	$\frac{1}{16}$	0	4	$\frac{4}{16}$
1	$\frac{4}{16}$	$\frac{4}{16}$	1	$\frac{4}{16}$
2	$\frac{6}{16}$	$\frac{12}{16}$	0	0
3	$\frac{4}{16}$	$\frac{12}{16}$	1	$\frac{4}{16}$
4	$\frac{1}{16}$	$\frac{4}{16}$	4	$\frac{4}{16}$
Sum	1	2	—	1

THE VARIANCE AS AN EXPECTED VALUE

The variance is the weighted mean of $(x-\mu)^2$, weighted by the probabilities. Thus the variance is the expected value of $(x-\mu)^2$,

$$\operatorname{Var}(X) = \mathrm{E}\big((X-\mu)^2\big).$$

This is a very useful form of the variance formula, because it will apply also to the continuous probability distributions introduced in Year 12.

TUTOR DEMO **An octahedral die**

An octahedral die has its eight faces numbered 1 to 8. It is thrown, producing a uniform distribution. Find the mean and variance of the distribution using the formula $\operatorname{Var}(X) = \mathrm{E}\big((X-\mu)^2\big)$.

YOUR TURN **Same method**

A tetrahedral die has its four faces numbered 1 to 4. It is thrown, producing a uniform distribution. Find the mean and variance of the distribution using the formula $\operatorname{Var}(X) = \mathrm{E}\big((X-\mu)^2\big)$.

WORKED EXAMPLE 5 **Keep for homework**

A die is thrown, producing a uniform distribution. Find the mean and variance of the distribution using the formula $\mathbf{Var}(X) = \mathbf{E}\big((X-\mu)^2\big)$.

SOLUTION

x	1	2	3	4	5	6	Sum	
$p(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	1	(a check)
$xp(x)$	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	$\frac{6}{6}$	$3\frac{1}{2}$	(the mean μ)
$(x-\mu)^2$	$\frac{25}{4}$	$\frac{9}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{9}{4}$	$\frac{25}{4}$	—	
$(x-\mu)^2p(x)$	$\frac{25}{24}$	$\frac{9}{24}$	$\frac{1}{24}$	$\frac{1}{24}$	$\frac{9}{24}$	$\frac{25}{24}$	$\frac{70}{24}$	(the variance)

Hence the mean is $3\frac{1}{2}$ and the variance is $\frac{70}{24} = 2\frac{11}{12}$.

CONCEPT 2 · the variance, alternative formula

THE VARIANCE: ALTERNATIVE FORMULA

The method of calculation in the four-tossed-coins table above worked well because the mean was a whole number. Things became a little turgid in the subsequent throwing-a-die experiment in Worked example 5, because the mean $3\frac{1}{2}$ was a fraction. If the denominator had been a larger number such as 287 rather than 2, things would have looked a great deal messier.

Fortunately, there is another formula for variance, again using weighted means, and this alternative formula makes calculations much easier:

$$\mathbf{Var}(X) = \sum x^2 p(x) - \mu^2.$$

This formula uses the squares of the values, and takes the weighted mean of these squares, weighted as always by the probabilities. Thus, like the first formula, it can be written in terms of expected values, this time in terms of the expected value of X^2,

$$\mathbf{Var}(X) = \mathbf{E}(X^2) - \mu^2.$$

THE FOUR COINS AGAIN, USING THE ALTERNATIVE FORMULA

The alternative formula makes the calculations more straightforward, whether or not μ is a whole number. The following table uses this alternative formula to recalculate the variance for the distribution obtained by throwing four coins and counting the heads.

x	0	1	2	3	4	Sum	
$p(x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$	1	(a check)
$xp(x)$	0	$\frac{4}{16}$	$\frac{12}{16}$	$\frac{12}{16}$	$\frac{4}{16}$	2	(the mean μ)
$x^2p(x)$	0	$\frac{4}{16}$	$\frac{24}{16}$	$\frac{36}{16}$	$\frac{16}{16}$	5	(this is $\mathrm{E}(X^2)$)

From the third row, $\mathrm{E}(X) = 2$, which is μ.

From the last row, $\begin{aligned}\mathrm{Var}(X) &= \mathrm{E}(X^2) - \mu^2 \\ &= 5 - 2^2 \\ &= 1, \text{ as we obtained before.}\end{aligned}$

7 VARIANCE OF A DISCRETE PROBABILITY DISTRIBUTION

- The *variance* of a discrete probability distribution with numeric data is
  $\mathrm{Var}(X) = \mathrm{E}\big((X-\mu)^2\big)$ or equivalently $\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2$.
- Written in terms of weighted means,
  $\mathrm{Var}(X) = \sum (x-\mu)^2 p(x)$ or $\sum x^2 p(x) - \mu^2$
  where the sums are taken over all the values in the distribution.
- The first of each pair gives the best intuitive understanding of the variance.
- The second of each pair is usually easier for calculation.
- Each difference $x - \mu$ is called the *deviation* of x from the mean μ.

EXTENSION: PROOF OF THE ALTERNATIVE FORMULA

The proof looks complicated only because of sigma notation. To prove the alternative formula for $\mathrm{Var}(X)$, start with the first formula, and move to the second.

$$\begin{aligned}\mathrm{Var}(X) &= \sum (x-\mu)^2 p(x) \\ &= \sum (x^2 - 2\mu x + \mu^2) p(x) && \text{(expand the square)} \\ &= \sum x^2 p(x) - \sum 2\mu x p(x) + \sum \mu^2 p(x) && \text{(expand the brackets)} \\ &= \sum x^2 p(x) - 2\mu \sum x p(x) + \mu^2 \sum p(x) && \text{(take out common factors)}\end{aligned}$$

Using the fact that $\sum p(x) = 1$ and $\sum x p(x) = \mu$,

$$\begin{aligned}\mathrm{Var}(X) &= \sum x^2 p(x) - 2\mu^2 + \mu^2 \\ &= \sum x^2 p(x) - \mu^2.\end{aligned}$$

TUTOR DEMO **A die with repeated numbers**

A die has its six faces numbered 1, 2, 2, 3, 3, 3. It is thrown, and X is the number on the top face. Find the mean and variance of the distribution using the alternative formula $\mathbf{Var}(X) = \mathbf{E}(X^2) - \mu^2$.

YOUR TURN **Same method**

A die has its six faces numbered 1, 1, 2, 2, 2, 4. It is thrown, and X is the number on the top face. Find the mean and variance of the distribution using the alternative formula $\mathbf{Var}(X) = \mathbf{E}(X^2) - \mu^2$.

WORKED EXAMPLE 6 **Keep for homework**

A die is thrown, producing a uniform distribution. Find the mean and variance of the distribution using the alternative formula $\text{Var}(X) = \text{E}(X^2) - \mu^2$.

SOLUTION

x	1	2	3	4	5	6	Sum	
$p(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	1	(a check)
$xp(x)$	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	$\frac{6}{6}$	$3\frac{1}{2}$	(the mean μ)
$x^2p(x)$	$\frac{1}{6}$	$\frac{4}{6}$	$\frac{9}{6}$	$\frac{16}{6}$	$\frac{25}{6}$	$\frac{36}{6}$	$\frac{91}{6}$	(this is $\text{E}(X^2)$)

From the third row, $\text{E}(X) = 3\frac{1}{2}$.

From the last row,
$$\begin{aligned}\text{Var}(X) &= \text{E}(X^2) - \mu^2 \\ &= \tfrac{91}{6} - \tfrac{49}{4} \\ &= 15\tfrac{1}{6} - 12\tfrac{1}{4} \\ &= 2\tfrac{11}{12}, \text{ as we obtained before.}\end{aligned}$$

CONCEPT 3 · standard deviation

STANDARD DEVIATION

The units of variance are the square of whatever units the values have. The *standard deviation* is the square root of the variance, and therefore has the same units as the values. Its usual pronumeral is σ, the lower-case Greek letter sigma (Latin 's'), standing for 'standard',

$$\sigma = \sqrt{\text{Var}(X)} \quad \text{or equivalently} \quad \sigma^2 = \text{Var}(X)$$

In the four-coins example at the start of this section, the units of the values and of the standard deviation are heads, and in Worked example 7 below, their units are dollars.

The term *standard deviation* implies that the deviations from the mean have been taken into account across all values of the distribution.

The other nice thing about the standard deviation is its proportionality. Standard deviation, like variance, is a measure of spread. If we stretch out all the values by doubling them, then we correspondingly double the standard deviation. If we spread out all the values by a factor of k, the standard deviation is multiplied by k. This will be proven in the Enrichment section of Exercise 13C (Questions 13 and 14).

8 STANDARD DEVIATION OF A DISCRETE PROBABILITY DISTRIBUTION

- The *standard deviation* is the square root of the variance. It is usually assigned the pronumeral σ. The Greek letter sigma stands for 'standard'.

  $$\sigma = \sqrt{\text{Var}(X)} \quad \text{or equivalently} \quad \sigma^2 = \text{Var}(X)$$

- The pronumeral for variance is thus σ^2.
- The standard deviation has the same units as the values.
- The variance and the standard deviation are *measures of spread*.

TUTOR DEMO **The pickpocket**

Sixteen friends meet at a café. One has $300 cash, three have $120 cash, two have $90 cash, four have $40 cash, and two have $20 cash. The other four have no cash. A pickpocket chooses one of the friends at random and takes all of his or her cash. Find the mean and the standard deviation of the amount the pickpocket takes.

YOUR TURN **Same method**

Ten friends are on a bus. Two have $50 cash, three have $30 cash, and one has $20 cash. The other four have no cash. A thief chooses one of the friends at random and takes all of his or her cash. Find the mean and the standard deviation of the amount the thief takes.

WORKED EXAMPLE 7 **Keep for homework**

Find the standard deviation of the distribution in Worked example 3 of Lesson 13B (the restaurant robbery).

SOLUTION

We use the alternative formula for variance, then take the square root.

x	0	40	50	80	180	350	560	Sum
$p(x)$	$\frac{5}{20}$	$\frac{1}{20}$	$\frac{4}{20}$	$\frac{3}{20}$	$\frac{2}{20}$	$\frac{4}{20}$	$\frac{1}{20}$	1
$xp(x)$	0	2	10	12	18	70	28	140
$x^2p(x)$	0	80	500	960	3240	$\mathbf{24\,500}$	$\mathbf{15\,680}$	$\mathbf{44\,960}$

From the last row,

$$\begin{aligned}\text{Var}(X) &= \text{E}(X^2) - \mu^2 \\ &= 44\,960 - 140^2 \\ &= 25\,360\end{aligned}$$

Taking the square root,

$$\begin{aligned}\sigma &= \sqrt{\text{Var}(X)} \\ &\doteqdot 159.25\end{aligned}$$

giving a spread of $\$159.25$ about the mean of $\$140$.

TEXTBOOK HOMEWORK · Exercise 13C

FOUNDATION

1 Consider a random variable X whose probability distribution is given in the table to the right.

x	1	2	3	4	Sum
$p(x)$	0.3	0.5	0.1	0.1	
$xp(x)$					
$(x-\mu)^2$					—
$(x-\mu)^2p(x)$					

a Copy and complete the table to calculate the mean $\mathrm{E}(X) = \mu$ and the variance $\mathrm{Var}(X)$ using the definition

$$\mathrm{Var}(X) = \mathrm{E}\big((X-\mu)^2\big).$$

b Calculate the standard deviation

$$\sigma = \sqrt{\mathrm{Var}(X)}.$$

2 This question uses the alternative formula for $\mathrm{Var}(X)$, rather than the definition, to calculate the variance for the random variable in the previous question.

x	1	2	3	4	Sum
$p(x)$	0.3	0.5	0.1	0.1	
$xp(x)$					
x^2					—
$x^2p(x)$					

a Copy and complete the table.

b Now calculate the variance using the alternative formula

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2.$$

3 For each random variable, calculate $\mu = \mathbf{E}(X)$. Then calculate the variance $\mathbf{Var}(X)$ twice, first using the definition $\mathbf{Var}(X) = \mathbf{E}\big((X-\mu)^2\big)$, then using the alternative formula $\mathbf{Var}(X) = \mathbf{E}(X^2) - \mu^2$. Use columns instead of rows in parts **c** and **d**.

a

x	0	1	2	3	4
$p(x)$	0.2	0.2	0.2	0.2	0.2

b

x	0	1	2	3	4
$p(x)$	0.0	0.1	0.2	0.3	0.4

c

x	$p(x)$
-2	0.3
-1	0.1
0	0.2
1	0.1
2	0.3

d

x	$p(x)$
1	0.1
2	0.4
3	0.2
4	0.2
5	0.1

DEVELOPMENT

4 **a** Calculate the expected value, variance and standard deviation for each random variable. For the variance, choose whether to use definition $\mathbf{Var}(X) = \mathbf{E}\left((X-\mu)^2\right)$ or the alternative formula $\mathbf{Var}(X) = \mathbf{E}(X^2) - \mu^2$.

(i)

y	0	1	2	3	4
$P(Y=y)$	0	0.5	0	0.5	0

(ii)

z	0	1	2	3	4
$P(Z=z)$	0.5	0	0	0	0.5

(iii)

v	$P(V=v)$
0	0.3
1	0.5
2	0.1
3	0.1
4	0

(iv)

w	$P(W=w)$
0	0
1	0.1
2	0.1
3	0.5
4	0.3

b Use the idea that the expected value measures the centre of the data, and the variance and standard deviation measure its spread, to comment on:
(i) how $\mathbf{E}(Y)$ and $\mathbf{Var}(Y)$ compare with $\mathbf{E}(Z)$ and $\mathbf{Var}(Z)$,
(ii) how $\mathbf{E}(V)$ and $\mathbf{Var}(V)$ compare with $\mathbf{E}(W)$ and $\mathbf{Var}(W)$.

5 A distribution that takes a single value is called *deterministic*, because it is no longer random. Our formulae for expected value and variance may still be calculated and the results are not a surprise, as this question demonstrates.
Calculate $\mathbf{E}(X)$ and $\mathbf{Var}(X) = \mathbf{E}\big((X-\mu)^2\big)$ for the distribution:

x	1	2	3	4	5
$P(X=x)$	0	1	0	0	0

6 John and Liam are keen basketballers and keep track of the number of baskets they score in games. Using the data from a large number of games, they have estimated the probability of scoring in any one game. Let the random variables J and L be the number of baskets scored by John and Liam respectively in a game. Their probability data is recorded in the tables below.

j	0	1	2	3	4
$P(J=j)$	0.35	0.2	0.1	0.25	0.1

ℓ	0	1	2	3	4
$P(L=\ell)$	0.2	0.3	0.4	0.1	0

a Calculate the expected value and variance for J and L using the alternative form $\mathbf{Var}(X) = \mathbf{E}(X^2) - \mu^2$.

b With reference to expected value, comment on who is the better player.

c With reference to variance, comment on who is the more consistent player.

7 The random variable X records the number on a spinner with sectors of equal size marked 1, 2, 3.

a What is the probability of each outcome, and what sort of distribution is this?

b Calculate the expected value $\mathbf{E}(X)$.

c Calculate the variance $\mathbf{Var}(X)$.

8 The *deviation of a score x from the mean* is often expressed in terms of how many standard deviations x lies from the mean μ. The formula for this is:

$$\textbf{number of standard deviations from the mean} = \frac{x-\mu}{\sigma},$$

where a negative sign means that the score x is below the mean.

a Englebert's score in his English test was 55. The test mean was $\mu = 65$ and the standard deviation was $\sigma = 5$. How many standard deviations was his score below the mean?

b Matthew's score in his Mathematics test was 54. The test mean was $\mu = 72$ and the standard deviation was $\sigma = 12$. How many standard deviations was his score below the mean?

c Comment on which score was more impressive, by noting which score was furthest from the mean in terms of the number of standard deviations.

9 Using the method of the previous question, that is, how many standard deviations a score is from the mean, decide which of each pair of test scores below is better.

a A score of 45 for Visual Arts (mean 60, standard deviation 15) or a score of 46 for Music (mean of 67, standard deviation of 12).

b A score of 88 for Earth Science (mean 70, standard deviation 9) or a score of 90 for Biology (mean of 75, standard deviation of 10).

c A score of 62 for Chinese (mean 50, standard deviation 6) or a score of 63 for Sanskrit (mean of 55, standard deviation of 4).

10 Jasmine is practising her accuracy with bow and arrow over 10 shots. The random variable X is the number of bull's-eyes obtained. She repeats this experiment twenty times and records the relative frequency as an estimate to the probability of each outcome. Her results are tabulated below.

x	0	1	2	3	4	5	6	7	8
$p(x)$	0	0.1	0.15	0.3	0.4	0	0	0	0.05

a Calculate the expected value and standard deviation for the experiment.

b An *outlier* is informally a value that is a long way away from the mean, and from the rest of the data, of the distribution. What value(s) would you think are outliers in this distribution?

c There are various quantitative ways of defining outliers. One such definition is *any value three or more standard deviations from the mean*. Are there any outliers using this definition?

d Jasmine realises that due to poor handwriting, her results have been wrongly interpreted. The corrected table is below. Recalculate the expected value and standard deviation for this new table.

x	0	1	2	3	4	5
$p(x)$	0	0.1	0.15	0.3	0.4	0.05

e Because of their distance from the mean, outliers can have a big influence on the value of the mean and standard deviation. Comment on the differences in the expected value and variance between the two tables above.

f Would it be valid in general to discard or 'correct' any outliers?

11 For some value of k, a random variable X with values 1, 2, 3, 4 is defined by

$$P(X = x) = kx, \quad \text{for } x = 1, 2, 3, 4.$$

Find k, then find the expected value and the standard deviation.

ENRICHMENT

12 A distribution is said to be *uniform* if every outcome has the same probability. Consider the random variable X of a uniform distribution with values $1, 2, \ldots, n$.

To complete this question you will find these formulae useful:

$$1 + 2 + 3 + \cdots + n = \tfrac{1}{2}n(n+1) \qquad 1 + 4 + 9 + \cdots + n^2 = \tfrac{1}{6}n(n+1)(2n+1)$$

a What is the probability $P(X = k)$ for integers $1 \leq k \leq n$?

b Calculate the expected value $\mathrm{E}(X)$.

c Calculate the variance $\mathrm{Var}(X)$.

d Compare your answers for expected value and variance with those for Question 7 involving a three-valued spinner, and with the example in the theory about throwing a standard six-sided die.

13 The expected value $\mathbf{E}(X)$ of a discrete probability distribution is μ.

a A constant a is added to all the values in the distribution. Let the random variable of this new distribution be Z, so that $Z = X + a$. Show that the expected value $\mathbf{E}(Z)$ of the new distribution is $\mu + a$.

b Each value in the distribution is multiplied by a constant k. Let the random variable of this new distribution be Z, so that $Z = kX$. Show that the expected value $\mathbf{E}(Z)$ of the new distribution is $k\mu$.

You have now proven that for all constants k and a,

$$\mathbf{E}(kX + a) = k\mathbf{E}(X) + a.$$

14 The mean and standard deviation of a discrete probability distribution are μ and σ. Use the formula $\mathrm{Var}(X) = \mathrm{E}\left((X-\mu)^2\right)$ to solve these two problems.

a A constant a is added to all the values in the distribution. Let the random variable of this new distribution be Z, so that $Z = X + a$. Show that the standard deviation $\sqrt{\mathrm{Var}(Z)}$ of the new distribution is also σ.

b Each value in the distribution is multiplied by a constant $k > 0$. Let the random variable of this new distribution be Z, so that $Z = kX$. Show that the standard deviation $\sqrt{\mathrm{Var}(Z)}$ of the new distribution is $k\sigma$.

You have now proven that for all constants $k > 0$ and a,

(standard deviation of $kX + a$) $= k\sigma$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 5 minutes **3 marks** Calculator allowed · show working · hints are below each question if you get stuck

These are the only school-paper items that need just this lesson's ideas. For more timed practice, do Questions 6 and 11 above in 15 minutes.

PP 1 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q1** 1 mark

The probability of success in a Bernoulli trial is 0.3. What is the variance?

(A) 0.09 (B) 0.7

(C) 0.3 (D) 0.21

Hint: a Bernoulli trial has two outcomes: success (the value 1) and failure (the value 0).

PP 2 **Baulkham Hills High School, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q12(a)(iii)** 1 mark

The following table is the distribution of a random variable X where $X \sim \text{Bin}(4, p)$.

x	0	1	2	3	4
$P(X = x)$	0.2401	a	b	0.0756	0.0081

In earlier parts of the question, it was found that $a = 0.4116$ and $b = 0.2646$.

(iii) Calculate $\text{Var}(X)$. 1

Hint: think about what the table now gives you.

PP 3 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q8** 1 mark

A Bernoulli variable, X, has a value of p such that $E(X) = 5Var(X)$.
Given that $p \neq 0$, what is the value of p?

(A) $\frac{1}{2}$ (B) $\frac{4}{5}$

(C) $\frac{1}{5}$ (D) $\frac{3}{5}$

Hint: a Bernoulli variable takes the value 1 with probability p and the value 0 otherwise, as in PP 1.

LESSON 13D

Sampling

A sample is the set of results when an experiment is performed a number of times. Its frequency table gives relative frequencies f_r, which correspond to the probabilities $p(x)$ of a probability distribution. The sample mean $\bar{x} = \sum x f_r$, the sample variance $s^2 = \sum x^2 f_r - \bar{x}^2$ and the sample standard deviation s are calculated in a table, exactly as μ, σ^2 and σ were. This lesson compares a sample with a theoretical distribution, explains census and survey, looks at samples where there are no exact probabilities, and ends with the correction factor for the sample variance.

CONCEPT 1 · sampling a theoretical probability distribution

SAMPLING

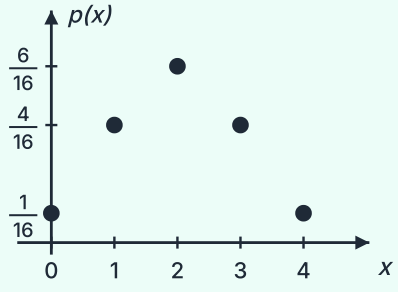
The word *sampling* is used when we perform an experiment a number of times and record the results. The set of results is a *sample*. One example of sampling is the large numbers of surveys from researchers, the government, and marketers. These surveys are constantly asking what we think about some product or service, how many people live in our house, whether we agree with the latest government policy, how many holidays we intend to take in the next five years, and so forth. The purpose of most of these surveys is to gain some idea of probabilities, usually so that future behaviours can be predicted.

Now that discrete probability distributions, expected value and variance have been introduced, this section discusses the relationship between sampling and probabilities. The subject is fundamental to statistics, and extremely complicated. Only a few very basic ideas are accessible here.

SAMPLING OF A THEORETICAL PROBABILITY DISTRIBUTION

We return once again to the experiment of throwing four coins and recording the number of heads.

x	0	1	2	3	4	Sum
$p(x)$	0.06	0.25	0.38	0.25	0.06	1
$xp(x)$	0	0.25	0.75	0.75	0.25	2
$x^2p(x)$	0	0.25	1.50	2.25	1.00	5



We concluded that $\mathrm{E}(X) = 2$ and $\mathrm{Var}(X) = 5 - 2^2 = 1$.

ROUNDING ERRORS AND SIMULATIONS

Each fraction in the earlier table has been rounded correct to two decimal places for comparison with the sample below: the last two rows were not recalculated. Notice that because of rounding errors, the first row may no longer add to 1, and the entry 1.50 in the middle of the bottom row is not $2^2 \times 0.38$. In this distribution, the errors could have been removed by rounding to four decimal places, but in general, rounding errors are a constant issue in statistics.

Can we confirm these probabilities and the subsequent calculations empirically? We could do a number of *simulations* or *trials*, which are independent *runs* of the experiment. It would be most unlikely that the resulting frequencies would be in the exact proportions of the theoretical table above, but it may be interesting to perform say 100 simulations and see what happens. The authors have done this using random numbers in an Excel spreadsheet rather than coins, because they only had three coins between them. Their results are in the worked example below.

9 SAMPLE DISTRIBUTION

Suppose that a number of independent *runs* (or *simulations* or *trials*) are performed of an experiment whose possible outcomes are discrete.

- The resulting *sample* then yields a *sample distribution* or *frequency table*.
- The relative frequencies f_r in the sample distribution correspond to the probabilities $p(x)$ in a probability distribution, and can be graphed the same way.

Now suppose also that the possible outcomes are numeric.

- The *mean* of the sample is denoted by $\bar{x}$, and corresponds to the expected value of the theoretical probability distribution:
  $$\bar{x} = \sum x f_r$$
- The *sample variance* s^2 and the *sample standard deviation* s are given by:
  $$s^2 = \sum (x - \bar{x})^2 f_r \quad \text{or} \quad s^2 = \sum x^2 f_r - \bar{x}^2$$
- The calculation of the mean and sample variance can be done in tabular form in the same way as the expected value and variance of a probability distribution.

SAMPLE STATISTICS AND THEORETICAL STATISTICS

The sample mean $\bar{x}$ will seldom be the same as the theoretical mean μ, which is why different symbols must be used. Similarly, the sample standard deviation s will seldom be the same as the theoretical standard deviation σ.

TUTOR DEMO **Three coins**

Three coins are tossed, and X is the number of heads.

a Write down the probability distribution of X. Then find the theoretical mean μ and the theoretical variance σ^2.

b The experiment was simulated 100 times, with these results:

x	0	1	2	3	Sum
f	14	35	39	12	100

Find the relative frequencies, the sample mean $\bar{x}$, the sample variance s^2 and the sample standard deviation s.

c Compare the sample statistics with the theoretical ones.

YOUR TURN **Same method**

Two coins are tossed, and X is the number of heads.

a Write down the probability distribution of X. Then find the theoretical mean μ and the theoretical variance σ^2.

b The experiment was simulated 50 times, with these results:

x	0	1	2	Sum
f	11	27	12	50

Find the relative frequencies, the sample mean $\bar{x}$, the sample variance s^2 and the sample standard deviation s.

c Compare the sample statistics with the theoretical ones.

WORKED EXAMPLE **From the text: keep for homework**

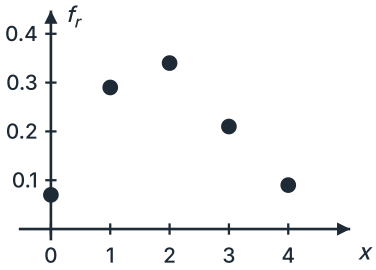
The authors simulated tossing four coins 100 times, and recorded the number x of heads each time. Here are their frequencies:

x	0	1	2	3	4	Sum
f	7	29	34	21	9	100

Find the relative frequencies and graph them. Then find the sample mean, the sample variance and the sample standard deviation, and compare them with the theoretical values.

SOLUTION

x	0	1	2	3	4	Sum
f	7	29	34	21	9	100
f_r	0.07	0.29	0.34	0.21	0.09	1
xf_r	0	0.29	0.68	0.63	0.36	$1.96 = \bar{x}$
$x^2 f_r$	0	0.29	1.36	1.89	1.44	4.98



The resulting *sample distribution* or *frequency table* has been structured as closely as possible to the theoretical table above. Its graph is drawn to the right above. In the table:

- The second line is the *frequency* f. There is no corresponding line in the probability distribution table.
- The third line is the *relative frequency* f_r, obtained by dividing each frequency by 100, which is the number of simulations performed. These relative frequencies correspond to the probabilities in the theoretical distribution. The graph of the sample distribution has been drawn, and should be compared with the theoretical graph in the box above.

Did you expect the relative frequencies to be closer to the probabilities, and the sample graph to be closer to the theoretical graph? That is a very difficult question, and beyond the course, but at least we can begin to see what may happen in practice.

- The fourth line gives us the *sample mean*, which is written as $\bar{x} = 1.96$. Placing a bar over the pronumeral x is the usual notation for the sample mean. It is the mean of the values weighted by their relative frequencies, just as the *theoretical mean* $\mu = 2$ (the expected value) is the mean of the values weighted by their probabilities.
- The last line gives us the mean of the squares of the values, again weighted by the relative frequency, corresponding to $\mathrm{E}(X^2)$ in the probability distribution table. It allows us to calculate the *sample variance* s^2 and the *sample standard deviation* s in exactly the same way that the *theoretical variance* σ^2 and the *theoretical standard deviation* σ were calculated:

$$\begin{aligned} s^2 &= \sum x^2 f_r - \bar{x}^2 \\ &= 4.98 - (1.96)^2 \\ &\doteqdot 1.14 \quad \textbf{(compare with 1)} \end{aligned}$$

$$\begin{aligned} s &= \sqrt{s^2} \\ &\doteqdot 1.07 \text{ heads} \quad \textbf{(compare with 1)} \end{aligned}$$

Although this course does not have the machinery to characterise whether the sample distribution differs significantly from the theoretical distribution, at least we have the basic data to begin the comparison.

CONCEPT 2 · populations, census and survey

POPULATIONS: CENSUS AND SURVEY

A school asks all its 1000 students to report how many brothers and sisters the student has. This is a *census*, because everyone is asked. Another similar-sized school chooses 100 students at random and asks them the same question. This is a *survey* or *sample* because not everyone is asked. When a marketer or the government wants to know something about the Australian population, almost always a survey is used because it is quicker and much cheaper than a census. Every five years, however, the Federal Government performs a compulsory census of the whole Australian population, and the results are eagerly awaited and discussed.

In any one census question, the relative frequencies are exact probabilities of a particular result when a person or thing is chosen at random. A *population distribution* is therefore akin to a probability distribution that has been calculated exactly using the laws of probability, so the pronumerals μ and σ are assigned to its mean and standard deviation.

When designing a survey, it is vitally important to know whether the sample is truly random. For example, many websites offer internet surveys, but such a sample is in no way random, because it includes only people visiting that website.

Also, the people chosen in the survey may be a significant proportion of the population. In the school survey above, 100 students were chosen out of only 1000. That should be taken into account, but it is beyond the present course.

Everyone wants to use surveys to approximate the statistical parameters of a population, but there are many problems, as Worked example 8 shows.

10 POPULATION: CENSUS AND SURVEY

- A *census* performs an experiment (perhaps by asking a question) on everyone or everything in a population. A *survey* samples only some of the population.
- The relative frequencies in a population distribution are the exact probabilities of a particular result when a person or thing is chosen at random from the population.
- The population mean is therefore assigned the pronumeral μ, and the population standard deviation is assigned the pronumeral σ.

TUTOR DEMO **Brothers and sisters**

A school of 800 students chooses 50 students at random, and asks each of them how many brothers and sisters he or she has. Here are the results:

x	0	1	2	3	4	5	Sum
f	8	17	15	6	3	1	50

a Is this a census or a survey?

b Find the relative frequencies, the sample mean $\bar{x}$ and the sample standard deviation s.

c Why will $\bar{x}$ seldom be exactly the mean μ for the whole school?

YOUR TURN **Same method**

A website asks its visitors how many pets they have at home. Forty visitors answer, with these results:

x	0	1	2	3	4	Sum
f	10	14	9	5	2	40

a Is this a census or a survey?

b Find the relative frequencies, the sample mean $\bar{x}$ and the sample standard deviation s.

c Is this a random sample of the Australian population? Explain.

WORKED EXAMPLE 8 **Keep for homework**

In 1992, astronomers confirmed the first discovery of an *exoplanet*, which is a planet orbiting a star other than our Sun. They soon found examples of multiple exoplanets orbiting the same star. Up to 28th November 2017, the website http://www.openexoplanetcatalogue.com reported that 3653 planets had been discovered, in 2657 nearby star systems in our galaxy, including our Sun. Here is the sample distribution for the number of stars that have x known planets orbiting them. Multiple-star systems are excluded because they are hard to classify, and only stars with at least one discovered planet are included. The frequency table below thus involves 3457 known planets orbiting 2527 single stars.

x	1	2	3	4	5	6	7	8	9	Sum
f	1921	401	133	47	14	5	3	1	2	2527 suns
xf	1921	802	399	188	70	30	21	8	18	3457 planets
f_r	0.7602	0.1587	0.0526	0.0186	0.0055	0.0020	0.0012	0.0004	0.0008	1
xf_r	0.7602	0.3174	0.1579	0.0744	0.0277	0.0119	0.0083	0.0032	0.0071	1.3680
x^2f_r	0.7602	0.6347	0.4737	0.2976	0.1385	0.0712	0.0582	0.0253	0.0641	2.5235

$$\bar{x} = \sum xf_r \\ \fallingdotseq 1.37 \text{ planets}$$

$$s^2 = \sum x^2 f_r - \bar{x}^2 \\ = 2.5235 - 1.3680^2 \\ \fallingdotseq 0.65$$

$$s = \sqrt{s^2} \\ \fallingdotseq 0.81 \text{ planets.}$$

This is a sample of the population of all the single-star-system stars in our Milky Way galaxy. Why would scientists never use this table alone to infer the distribution of exoplanets across all the single-star systems in the galaxy?

SOLUTION

[Some of the more obvious reasons.]

- Only larger planets are counted because smaller planets are hard to find. The number of planets observed to orbit each star in the table will probably be less than the actual number of planets orbiting that star.
- We can't yet show that a star has no planets, so it would be most confusing to include the value 0.
- These stars are near Earth, so they may not represent the average situation in the galaxy. In particular, in denser regions, stars are closer, and planets may perhaps be more easily stripped away by near collisions with other stars.
- This is an extremely small sample of the 100–400 billion stars in the galaxy.

CONCEPT 3 · sampling when there are no exact probabilities

SAMPLING WHEN THERE ARE NO EXACT PROBABILITIES

Until now, almost everything has been based on equally likely possible outcomes. The four-coins probabilities were based on 16 equally likely outcomes when four coins are tossed. The number-of-exoplanets discussion was based on the unknown population of stars in the galaxy, with the equally likely possible outcomes being choosing a single-star-system star in the galaxy at random and counting its planets (not that we have any idea how such an experiment could be performed).

The first worked example below, about litters of mice, comes from the text.

TUTOR DEMO Goals in a season

A football team recorded the number x of goals it scored in each of its 40 matches last season:

x	0	1	2	3	4	5	Sum
f	6	12	11	7	3	1	40

a Find the sample mean $\bar{x}$ and the sample standard deviation s.

b The coach wants to use the table to predict the team's results next season. Give two questions that should be asked first.

YOUR TURN Same method

A bird watcher recorded the number x of eggs in each of 25 nests of one species of bird:

x	2	3	4	5	6	Sum
f	3	6	9	5	2	25

a Find the sample mean $\bar{x}$ and the sample standard deviation s.

b Are the relative frequencies the exact probabilities for a nest of this species? Explain.

WORKED EXAMPLE **From the text: keep for homework**

In 1903, a German researcher recorded the number x of baby mice in each litter of mice in his laboratory (Yule & Kendall 1950, *An introduction to the theory of statistics*, Charles Griffin, page 121):

x	1	2	3	4	5	6	7	8	9	Sum
f	7	11	16	17	26	31	11	1	1	121 litters

Find the sample mean, the sample variance and the sample standard deviation. What would the relative frequencies mean as probabilities?

SOLUTION

x	1	2	3	4	5	6	7	8	9	Sum
f	7	11	16	17	26	31	11	1	1	121 litters
xf	7	22	48	68	130	186	77	8	9	555 babies
f_r	$\frac{7}{121}$	$\frac{11}{121}$	$\frac{16}{121}$	$\frac{17}{121}$	$\frac{26}{121}$	$\frac{31}{121}$	$\frac{11}{121}$	$\frac{1}{121}$	$\frac{1}{121}$	1
xf_r	$\frac{7}{121}$	$\frac{22}{121}$	$\frac{48}{121}$	$\frac{68}{121}$	$\frac{130}{121}$	$\frac{186}{121}$	$\frac{77}{121}$	$\frac{8}{121}$	$\frac{9}{121}$	$\frac{555}{121} = \bar{x}$
x^2f_r	$\frac{7}{121}$	$\frac{44}{121}$	$\frac{144}{121}$	$\frac{272}{121}$	$\frac{650}{121}$	$\frac{1116}{121}$	$\frac{539}{121}$	$\frac{64}{121}$	$\frac{81}{121}$	$\frac{2917}{121}$

$$\bar{x} = \sum xf_r = \frac{555}{121} \div 4.59 \text{ baby mice}$$

$$s^2 = \sum x^2 f_r - \bar{x}^2 = \frac{2917}{121} - \frac{308\,025}{14\,641} \div 3.0689$$

$$s = \sqrt{s^2} \div 1.75 \text{ baby mice.}$$

The sample statistics were easily calculated. The obvious intention, however, is to talk about all mice, and this time there are no theoretical probabilities based on equally likely possible outcomes, and no population. We would normally interpret the relative frequencies as probabilities that a mouse litter, chosen at random, has this many baby mice, but what would such a probability mean?

- Is such a probability a statement about the physical world, in this case mice?
- Or is it a statement about our lack of knowledge about the size of a litter?

In the second case, the probability is never fixed, but changes all the time as further information about mouse litters comes in.

Then there are also the usual questions about the experiment. For example:

- Are these laboratory mice better fed and housed than wild mice?
- Is there some subtle genetic difference between these mice and other mice?
- Does weather affect litter size, and were weather patterns over the last few years normal, whatever that means?

PREDICTING THE FUTURE

The most important, and the most tricky, use of sampling is to predict the future. The worked example below, about rainy days in April, comes from the text.

WORKED EXAMPLE **From the text: keep for homework**

From the website http://www.bom.gov.au, here are the numbers of years from 1913 to 2012 when there were x rainy days in April, recorded at Observatory Hill, Sydney (where 'rainy day' means here a day when the rainfall was greater than zero.)

x	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	Sum
f	1	1	3	2	7	9	3	9	10	9	4	9	2	4	9	7	5	2	1	2	1	100

Find the sample mean, the sample variance and the sample standard deviation. Can the table be used to predict the number of rainy days in April in future years?

SOLUTION

x	3	4	5	6	7	8	9	10	11	12	13
f	1	1	3	2	7	9	3	9	10	9	4
f_r	0.01	0.01	0.03	0.02	0.07	0.09	0.03	0.09	0.10	0.09	0.04
xf_r	0.03	0.04	0.15	0.12	0.49	0.72	0.27	0.90	1.10	1.08	0.52
x^2f_r	0.09	0.16	0.75	0.72	3.43	5.76	2.43	9.00	12.10	12.96	6.76

x	14	15	16	17	18	19	20	21	22	23	Sum
f	9	2	4	9	7	5	2	1	2	1	100
f_r	0.09	0.02	0.04	0.09	0.07	0.05	0.02	0.01	0.02	0.01	1
xf_r	1.26	0.30	0.64	1.53	1.26	0.95	0.40	0.21	0.44	0.23	12.64
x^2f_r	17.64	4.50	10.24	26.01	22.68	18.05	8.00	4.41	9.68	5.29	180.66

$$\bar{x} = \sum xf_r = 12.64 \text{ rainy days}$$

$$s^2 = \sum x^2 f_r - \bar{x}^2 = 180.66 - 12.64^2 \div 20.89$$

$$s = \sqrt{s^2} \div 4.57 \text{ rainy days.}$$

In the five subsequent years from 2013 to 2017, the successive numbers of rainy days in April were 16, 15, 23, 13 and 10. Here are some questions that the reader may raise, but the course cannot answer.

- Are these subsequent results consistent with the probabilities suggested by the relative frequencies in the table?
- Droughts last for many years, so perhaps rainy Aprils come in groups? If there are cycles in weather with periods of say 15 years, or 1100 years, or 125 000 years, then how could we use this table as a predictor of future weather? Are the last five years' results consistent with a permanent and unprecedented shift in climate?

- What does the probability of 10 rainy days in April actually mean? Is it a statement about the physical world, in this case the climate, or is it statement about our lack of knowledge? If such probabilities exist, do they vary over time, in which case how does one define them? There are certainly no equally likely possible outcomes in sight here to base a theoretical approach on.

Clearly a single sample distribution such as this does nothing to answer the urgent questions people are asking. Far more data on all sorts of different phenomena, together with sophisticated statistical tests that can quantify exceptional results and take into account possible cyclical data, are required before any reliable conclusions can be made. Climate science is extremely complicated.

CONCEPT 4 · a correction factor for the sample variance

EXTENSION: A CORRECTION FACTOR FOR THE SAMPLE VARIANCE

When we know the theoretical mean μ and we are sampling to find the variance, there is no problem with the formulae for the sample variance. When, however, we are sampling both to find the mean and to find the variance, then the sample mean will drift very slightly towards the sample results, with the effect that the sample variance will tend to be slightly smaller than it should be.

This phenomenon is much discussed, and the standard answer is that the sample variance should be multiplied by a correction factor $\frac{n}{n-1}$, where n is the size of the sample.

Thus in the worked example above about rainy days in April, there were 100 results, and we were using a sample mean rather than a theoretical mean, so the correction factor is $\frac{100}{99}$. The worked example below uses it.

The larger the size n of the sample, the less difference the correction makes.

Your calculator may have a button $\boxed{\sigma_n}$ or something equivalent. This gives the standard deviation without the correction factor, and is all that is required in this course. It may also have a button labelled $\boxed{\sigma_{n-1}}$ or equivalent, which applies the correction factor. This button is not required in this course.

TUTOR DEMO **Correcting a sample variance**

A sample of 40 football matches gave a sample mean of 1.8 goals and a sample variance $s^2 = 1.56$. The sample mean was used in finding s^2, because the theoretical mean is not known. Apply the correction factor to find the corrected variance and standard deviation, and compare them with s^2 and s.

YOUR TURN **Same method**

A sample of 16 values gave a sample variance $s^2 = 3.75$. The sample mean was used in finding s^2, because the theoretical mean is not known. Apply the correction factor to find the corrected variance and standard deviation, and compare them with s^2 and s.

WORKED EXAMPLE **From the text: keep for homework**

Apply the correction factor to the sample of rainy days in April in the worked example above.

SOLUTION

There were 100 results, so the correction factor is $\frac{100}{99}$. Using the correction factor would yield

$$\bar{x} = 12.64 \text{ rainy days (as before)} \qquad s^2 = 20.89 \times \frac{100}{99} \div 21.10 \qquad s = \sqrt{s^2} \div 4.59 \text{ rainy days.}$$

TEXTBOOK HOMEWORK · Exercise 13D

APPROACHING THE FOLLOWING EXERCISE

The following exercise contains many suggestions for experiments that will generate data. In each situation you should be asking a number of important questions:

- How can you ensure that the data are randomly generated?
- Are there any biases due to the population used for your experiment? For example, measuring the heights of students in the class will give results different from measuring the heights of students in the school.
- How can you complete the experiment efficiently and accurately, including recording the data? Tally tables, and accurate record-keeping, are two important skills in experiments.
- What special equipment is needed, such as stopwatches, tape measures, metre rulers?
- How big does the sample size need to be to get a reasonable estimate? This might become clearer as the experiment proceeds, depending on whether the data appear stable.
- Is your data from a population, with mean μ and standard deviation σ, or from a sample, with sample mean $\bar{x}$ and sample standard deviation s?
- What are the mean and standard deviation measuring, and what is their interpretation in the context of the experiment? Are there theoretical or population means and standard deviations to compare the data with? If so, what conclusions can you draw?

USING A SPREADSHEET

The probabilities of a distribution need to be calculated, and the frequencies of a sample need to be tallied. Once this is done, however, the calculations of mean, variance and standard deviation can be automated on a spreadsheet, provided that you know how to do two things:

Write formulae into cells and Fill down and fill right.

You are invited to use a spreadsheet in this exercise to save time in the calculations. Once a calculation has been set up, it can be copied and adapted.

Many simulations can be done using *random numbers*: see Question 5, for example, and see the table of random numbers after Exercise 13D. Excel has two very useful random number functions:

`RANDBETWEEN (a, b)` generates a random integer in the interval $a \leq x \leq b$. For example, `RANDBETWEEN (11, 20)` may generate 14, or 20, or 11, or 15.

`RAND ()` generates a random number in the interval $0 < x < 1$. For example, it may generate 0.865348256572775, or 0.44037476507047, or 0.0174096864227915.

Once you have a random number in a cell, the commands fill down (`Ctrl+D`) and fill right (`Ctrl+R`) will give you as many random numbers as you like.

This is not a normal exercise with a normal set of questions. Apart from Question 1, it is a set of suggestions for activities and investigations, some of which could easily be turned into projects. Most questions ask for a number of simulations (or trials or runs) of an experiment in order to create a sample, followed by an analysis of the results.

First do Question 1. After that, there is no expectation that anyone should simply work through the exercise, because the subsequent questions all take some time, however they are done. They are intended to be adapted to the needs of the class or the individual.

These simulations could be done by individual students, but they could also be done in groups, or by the class as a whole. Most could also be done on a spreadsheet using random numbers. This requires setting up a spreadsheet, which also takes time.

FOUNDATION

1 Three dice are thrown. Let X be the number of dice showing a 5 or 6.

a The probability distribution for this experiment is shown below (it can easily be calculated using a probability tree diagram).

x	0	1	2	3
$p(x)$	$\frac{8}{27}$	$\frac{12}{27}$	$\frac{6}{27}$	$\frac{1}{27}$

Graph the distribution. Then calculate the theoretical mean μ, the theoretical variance σ^2, and the theoretical standard deviation σ.

b The frequency table below gives the results when the experiment was done 100 times:

x	0	1	2	3	Sum
f	33	47	16	4	100

Calculate the relative frequencies, and graph them. Then calculate the sample mean $\bar{x}$, the sample variance s^2, and the sample standard deviation s.

c Do the sample results appear to be consistent with the theoretical results?

2 Two dice are thrown. Let X be the sum of the two numbers on the dice.

a Here is the theoretical probability distribution of the experiment, together with the calculations for obtaining the mean and variance. From the table, write down the mean μ, the variance σ^2, and the standard deviation σ.

x	2	3	4	5	6	7	8	9	10	11	12	Sum
$p(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	1
$xp(x)$	$\frac{2}{36}$	$\frac{6}{36}$	$\frac{12}{36}$	$\frac{20}{36}$	$\frac{30}{36}$	$\frac{42}{36}$	$\frac{40}{36}$	$\frac{36}{36}$	$\frac{30}{36}$	$\frac{22}{36}$	$\frac{12}{36}$	7
$(x-\mu)^2$	25	16	9	4	1	0	1	4	9	16	25	—
$(x-\mu)^2p(x)$	$\frac{25}{36}$	$\frac{32}{36}$	$\frac{27}{36}$	$\frac{16}{36}$	$\frac{5}{36}$	0	$\frac{5}{36}$	$\frac{16}{36}$	$\frac{27}{36}$	$\frac{32}{36}$	$\frac{25}{36}$	$\frac{210}{36}$

b Throw a pair of dice 72 times and record your sample distribution in a relative frequency table.

c How do your probabilities in part **a** compare with the relative frequencies obtained in your experiment? Draw the two graphs and compare them.

d Use the sample distribution table to calculate the sample mean $\bar{x}$, the sample variance s^2, and the sample standard deviation s. The sample mean will not be a whole number, so use the alternative formula for variance.

e How do the sample statistics $\bar{x}$ and s compare with the theoretical statistics μ and σ?

f To improve your estimation, combine your results with those from other members of your class.

3 Design some censuses in your class. Draw up a population distribution table for each, and calculate the population mean μ, the population variance σ^2, and the population standard deviation σ. Here are some suggestions.

a The month number of each student's birthday.

b The number of brothers and sisters of each student.

c Each student's writing speed: count the number of times in one minute that a student can fully write the four words, 'The quick brown fox'.

d The number of sports they do, or the number of musical instruments they play.

e The number of pets at home. If they have goldfish, decide whether that result is an outlier or should be rejected.

f How many days in the last week did each student cook a meal?
Why is, or is not, each class census an unbiased sample of the school population?

4 Many experiments in this chapter have involved throwing a die. It is important to know *how random* the results will be that are obtained from throwing a die.

a Throw a die 40 times and write down the results one after the other.

b Are the probabilities approximately *uniform*, that is, the same for each outcome?

c We now investigate whether each outcome on the die is *independent* of the previous roll. Perhaps the way you roll the die affects things here?

(i) Write down the 39 *differences* between successive rolls, discarding any minus sign.

(ii) Draw up a sample distribution table, and calculate the sample mean $\bar{x}$ and sample standard deviation s.

(iii) Question 10 of Exercise 13B asked for the probability distribution table of an experiment equivalent to this, namely 'Throw two dice and record their difference'. Copy and complete the table to find the standard deviation σ.

x	0	1	2	3	4	5	Sum
$p(x)$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$	1
$xp(x)$	0	$\frac{10}{36}$	$\frac{16}{36}$	$\frac{18}{36}$	$\frac{16}{36}$	$\frac{10}{36}$	$\frac{70}{36}$

(iv) Check whether your experimental probabilities agree with the theoretical results. Draw the two graphs and compare them.

5 Many calculators have a random-number generator. One calculator has a button labelled `Ran#` (returning a three-digit number between 0 and 1) and a button labelled `ranint ()` (for example, `ranint (1, 13)` returns a random integer between 1 and 13 inclusive). Use your manual, or ask others, to find what functions are available on your calculator.
Excel's `RAND` function is a random-number generator. Many websites provide random numbers. There is also a list of random numbers at the end of this exercise. If your method generates a three-digit number and only one digit is required, simply discard the unwanted digits.

a Generate a single-digit random number 50 times, keeping the numbers. Calculate the sample mean $\bar{x}$ and the sample standard deviation s. Compare them with the uniform distribution on the numbers $0, 1, \ldots, 9$, where $\mu = 4\frac{1}{2}$ and $\sigma^2 = 8\frac{1}{4}$ (these results were calculated in general in Question 12 of Exercise 13C).

b Discard the results of part **a** if the digit is 0 or 9, calculate the new sample mean $\bar{x}$ and sample standard deviation s, and compare them with the corresponding uniform distribution of the numbers $1, 2, \ldots, 8$, where $\mu = 4\frac{1}{2}$ and $\sigma^2 = 5\frac{1}{4}$. This models an eight-sided die used in some games.

DEVELOPMENT

6 This question models a simple lottery. You will be using random-number simulations to estimate the probabilities and payouts. Theoretical probabilities are not required.
Write down any four distinct single-digit numbers between 0 and 9 inclusive. These will be your winning numbers for the next four parts.

a Generate, by any method, four distinct random single-digit numbers. If you get a number that has already occurred, just discard it and generate a new number.

b Record the number of matches you have between your four random numbers and your four winning numbers. The number of matches is your random variable X.

c Repeat this experiment say 30 times and tabulate your results as a frequency table for X. These are your experimental estimates of the probability of each outcome.

d Calculate your expected payout if 4 matches wins you $\$100$ and 3 matches wins you $\$10$. Then calculate your expected profit or loss if entering the game costs $\$2$.
[A longer investigation]
You may change the range of numbers, the number of numbers, the payouts, and the cost of entering the game, to generate other results. How much of the procedures in parts **a–d** can you automate if you write your simulations in Excel?

7 [Benford's law]

This question models *Benford's law*, which is used, amongst other things, in forensic analysis of large data sets, for example to spot tax fraud. It models the frequency distribution of the leading digit in each score of the data by the following table:

x	1	2	3	4	5	6	7	8	9
$p(x)$	30.1%	17.6%	12.5%	9.7%	7.9%	6.7%	5.8%	5.1%	4.6%

a Copy and complete the table to find the mean μ and standard deviation σ.

b We can test Benford's law using the first digit of the powers of 2 from 2^0 to 2^{99}. Generate a random two-digit number d (whose leading digit may be 0). Define a random variable X to be the leading digit of 2^d. For example,

If $d = 36$, then $2^d \doteqdot 6.8719 \times 10^{10}$, so $X = 6$.

Repeat this experiment 40 times, record your results in a frequency table, and calculate the sample mean $\bar{x}$ and the sample standard deviation s. Do your results agree reasonably with Benford's law?

Applying Benford's law to internet data: Obtain a large set of numerical data, such as stock prices, house prices, population numbers, death rates, lengths of rivers. Choose data that varies widely in its range across several orders of magnitude: a distribution with range from say 10 to 100 would not be a good set of data for this experiment. See if the distribution of the leading digits in the data agrees reasonably with Benford's law.

A possible project: What formula generated these nine percentages in the table above? How can it be proven, and what assumptions are being made in that proof? What is the history of Benford's law? What are some typical situations, in all sorts of contrasting fields, where Benford's law is useful? In what situations does Benford's law begin to break down, or break down completely?

8 This experiment estimates throwing accuracy. First, you need an object that you can throw and will not bounce much, such as a small bag filled with sand, or tightly rolled paper. Secondly, find a clear wall that you can stand some distance away from. On the wall at ground-level, mark a bull's-eye that a thrower is to aim for. Use tape to place successive marks at 10 cm intervals from the bull's-eye, horizontally to the left and to the right, at the base of the wall. Label these marks $\ldots, -2, -1, 0, 1, 2, \ldots$. Later, if people are missing the bull's-eye badly, you may have to increase the number of marks. Thirdly, mark a position quite some distance away to throw from.
Throw the projectile 50 times and record the integer nearest to where the projectile lands. The successive throws can be done to test an individual's accuracy, or a group's accuracy. Calculate the sample mean and sample standard deviation of the experiment.
This experiment can also be done as a continuous probability experiment. If the results are graphed, the result might be expected to be the bell-shaped curve of the *normal distribution* that you will study in Year 12. A discrete version of this curve may be evident if you graph your results.

9 **a** Open a novel on a random page, and choose a word at random. Let X be the random variable recording the number of letters in the word.

b Compare results obtained from a 'light' novel and a more serious text or classic, such as you might study in English.

c Will your results be different if you use a textbook from history or science? Perhaps different groups in the class could use different types of books.

10 The distribution of letters in English is well documented and may be used in code cracking, as well as having application to games such as Scrabble.

a Use the internet to find the expected distribution of letters in an English document. (A search for 'distribution of letters in English' should return a suitable result.)

b Again using a random page in a novel, count the number of occurrences of each letter within a block of 200 letters, and hence make your own estimate of the frequencies of each letter.

c Do the results depend on the type of document selected?

11 Many irrational mathematical constants are called *normal*, meaning that the distribution of the digits in their decimal expansion is uniform, with every digit being equally likely. Use the internet to find the decimal expansions of constants such as π and $\sqrt{2}$, and find the frequency of each digit in their first 50 digits.

12 [Experiments involving estimation]
These experiments involve certain difficulties with randomisation that you will need to minimise in your experimental design.

a First read the list of experiments below and discuss what issues may occur.

b To be completed in groups or with the whole class:
(i) Draw a line on the board. Each member of the group is to estimate the length of the line, correct to the nearest centimetre.
(ii) Each member is asked to draw a 10 cm line on the board without a ruler. Then measure the answer correct to the nearest centimetre.
(iii) Each member is to draw a circle of radius 20 cm on the board. Then estimate its roundness by measuring the difference between the longest and shortest diagonal, correct to the nearest 1 cm.
Again, these experiments could also be done as continuous probability experiments.

13 **Applying the theory to internet data:** The internet is a vast source of data. Here are some suggestions of how some internet data can be investigated. They should prompt many more similar investigations.

a Most countries in the world have land borders with other countries. Let X be the number of adjacent countries of a country, and draw up a frequency table for X. Then comment on your results, with the help of people with a knowledge of geography.

You could do the same analysis of the states of one country, such as USA, Germany or Australia (decide first whether to regard the Northern Territory and ACT as states).

b There are less than 100 naturally occurring elements, but over 300 naturally occurring isotopes of these elements. Let X be the number of isotopes of an element, and draw up a frequency table for X, with mean and standard deviation. Then comment on your results, with the help of people with a knowledge of physics and chemistry.

You will soon run into huge problems of classification. What shall we class as 'naturally occurring'? Do we include radioactive isotopes that have extremely short half-lives? Do we include isotopes not found on Earth? Do we include isotopes and elements produced in nuclear reactors?

c Natural disasters are rich sources of data. Let X be the number of major earthquakes worldwide in a particular year. Draw up a frequency table for X, and comment on your results. You will need to define the term 'major earthquake', probably after looking at some data, and also decide whether several large earthquakes close together in time and space count as one earthquake or many. You may also like to consider smaller earthquakes and restrict the study to some region of the globe.

Major hurricanes could be the subject of a similar study.

ENRICHMENT

14 [The German tank problem]

The enemy has N tanks, each labelled with a unique serial number from 1 to N. Unfortunately the number of tanks is unknown, but that would be useful information to the war effort. Tanks are occasionally captured and their serial numbers read. Counter-intelligence operatives want to use these serial numbers to estimate the total number N of tanks.

They begin with the assumption that these serial numbers lie randomly within the interval from 1 to N. For the purposes of modelling this problem, we will work as if the serial numbers are equally spaced.

Suppose that k serial numbers have been discovered, with m being the largest of them. The case $k = 3$ and $m = 9$ is shown in the diagram.



a Explain why the average number of undiscovered serials between the k samples on the interval from 1 to m is $\dfrac{m-k}{k}$.

b If this same gap extends past the largest discovered serial number, explain why

$$N = m + \frac{m}{k} - 1.$$

c Test this estimate by modelling the tank problem.

(i) Working in pairs, the enemy partner chooses a two-digit number N, representing the number of tanks they have. They do not share this.

(ii) The enemy tank owner generates a random number from 1 to N and allows the counter-intelligence partner to 'discover' this serial number. The counter-intelligence operative tabulates the number k of discovered serials, the maximum serial number m discovered so far, and their estimate of N using the previous formula.

(iii) How quickly and accurately does the scheme estimate the correct value of N? How many tanks must be discovered before the estimate is within 10% of the correct value?

A TABLE OF RANDOM NUMBERS

Some cautions about using these or any table of random numbers:

- Any printed set of numbers such as these is only a simulation of random numbers. Once they are written down they are no longer random.
- If you are repeating an experiment, never take your random numbers from a place on this table where you have taken numbers before.
- It may be safer to search for 'table of random numbers' on the internet than to use these.
- Excel generates random numbers with the functions `RAND` and `RANDBETWEEN`, and most calculators generate them. How do they generate them?

37743	65520	55183	10968	38039	67295	66872	49849	24546	23324	77131	09038	02082	50408
41801	02262	85716	34590	09006	66392	21083	70211	01972	13007	29700	23977	26832	04955
79271	24018	05443	57916	79007	73955	33197	66964	99935	91957	61833	58130	20211	74807
07252	64966	95430	86966	82324	30105	53945	74630	54709	31824	17172	04797	29660	35607
44907	29330	57843	70710	05639	74016	68147	31718	71858	52629	25494	05112	40372	05998
13655	20830	77618	66061	43060	28829	89975	44934	83615	00648	19074	48971	22506	55971
62113	17825	47021	14189	17171	88516	52356	70962	53516	84486	74430	99783	06920	29668
47698	78539	99974	87544	87961	41074	87394	25210	00183	53751	26340	02438	11297	17016
71726	85738	28023	91959	96532	87421	19680	51877	37115	29660	25994	58836	63527	33556
89917	65236	74820	33196	15434	40370	19366	14720	50378	21987	40163	80380	94243	92858
45152	37604	30995	82334	02529	13058	86885	92913	61053	91709	13261	51796	95365	80887
94383	87457	76296	99349	80233	31202	66181	91339	44328	43728	39172	65651	66531	66694
26550	59696	40868	99275	87036	28556	08335	99728	02106	59054	14716	11957	96302	08814
93391	17072	82546	70322	73149	09950	32449	59990	11004	00805	71630	32220	50003	41740
00019	18462	02044	78320	37180	91282	55680	26126	92615	53419	08942	72322	61082	51852
27142	02218	42903	16862	25693	50829	89282	26090	77915	79441	72611	93606	47853	77171
81790	83966	48086	87531	42531	51278	22542	59990	62679	03263	54306	26536	02516	90318
98440	03241	70082	19577	39788	16819	41482	93977	75793	56193	06199	76316	24307	49053
37407	78320	56407	63424	87287	74701	54201	34224	07333	42224	93650	14828	93670	23765
52371	53632	57363	17378	42455	95610	19732	90354	68279	08886	32663	18284	09316	36288
23921	49125	27951	71912	67462	92257	02289	62690	78746	32550	83974	34946	35050	69424
80994	45961	20780	80912	65499	60536	97782	12724	06612	52555	00657	92307	28239	31544
57072	67909	91887	94239	33074	72642	04470	01144	03400	13543	51126	78094	46792	85639
53028	42821	66307	11004	50198	56513	34633	02830	09316	00376	64396	54481	56535	77200
03481	53562	35294	37918	55334	20976	89131	33172	24283	74992	69490	89872	85848	20303
68542	12761	82050	62092	76990	10975	41729	45090	66887	49608	25854	17644	99446	26275
97832	20775	57850	72669	18829	02244	05069	81133	68073	55611	66672	56212	88173	84300
38636	91223	55429	90662	20162	69815	75145	90034	71484	92532	01942	49805	96687	28373
85068	53478	51514	91358	38715	89730	76181	42089	37326	66477	53189	14313	35914	49938
90136	66483	15368	28691	27533	58321	52769	02744	71611	52382	16702	97590	52136	40617
87569	03275	05223	95931	02900	38457	81146	48751	27668	39470	77154	88585	49599	74388
95278	90584	99483	82082	25149	67110	96494	36290	51806	44290	18492	34675	88514	63587
97345	12886	34569	08356	39236	83625	58118	55409	69976	00707	03967	51579	05318	48122
26393	15280	46270	54100	46962	63071	99405	30169	37334	61219	22175	35222	27080	02257
93225	64579	36496	06554	07701	64046	34410	48872	97568	68980	20999	09930	76920	68352
35601	52966	12289	93810	89525	25301	17966	33739	23866	84899	76304	72323	92317	51934
83037	53612	68980	85346	17591	70582	81182	84410	38393	79195	67283	15293	03191	37464
59490	62620	14356	69518	91673	06038	89207	54660	32172	21772	55311	41429	41614	09193
70806	86669	55723	42381	69705	19586	93131	46682	32216	30809	88198	00277	80048	86458
48813	60741	96728	67204	39217	16605	72911	32330	87809	63919	76696	08795	19962	07818
48931	05297	15832	89078	23535	51333	93176	35684	72804	61267	15158	39390	77725	23384
23167	57283	23289	13961	94625	38343	18805	36323	25804	75008	43210	84912	39633	79735
15381	34341	18518	86564	89450	53466	62145	77678	75117	44440	70303	34968	82958	06781
50686	16595	94742	26433	51716	58551	58742	89828	20345	01223	10048	22948	84876	40955
15876	55389	62771	06611	45887	09424	93465	61135	67494	14062	96178	28119	66176	44113
52727	43359	90793	84253	15340	31422	23893	18080	39469	07207	77032	42184	41597	02507

11080	05202	82059	17571	81536	71887	72847	97671	08840	12168	10429	34657	73687	15209
33555	93526	37338	28479	06074	63982	86933	21808	24745	36996	05804	76451	52050	06376
68784	99154	74006	95318	62528	61211	82977	07981	30813	62531	85487	06040	44863	20697
35035	10254	98654	85034	89902	27024	29227	75663	95969	99961	62004	47571	94526	86443
81097	79671	55086	59182	24623	88864	33197	32783	05015	69100	53255	15310	54334	55445
16316	84845	94584	13138	32907	91051	33021	51660	80844	81737	82469	63789	88037	54421
06631	89667	40308	32886	90400	05865	14201	68751	35174	39438	34796	68027	79168	06994
58519	42683	42723	78754	80957	87208	98625	67616	69671	80173	67412	43213	22006	10792
37689	70982	47903	30488	11438	76221	77716	29452	74337	33157	95020	38207	91062	79001
94427	54174	25610	53018	64393	93225	86605	60584	95038	66257	97613	62254	10952	80745
92453	87817	29012	67312	35511	75367	41192	31286	67052	35359	22375	29264	73532	39672
63169	70677	82912	35678	27715	44349	84968	85296	36502	70061	76606	29060	74181	75115
32172	75178	22374	95893	98768	13599	17488	04423	16560	55070	33528	59941	64095	95204
89910	85449	28611	03537	84271	69284	01348	37595	50285	91677	97866	81955	40472	76316
03349	12306	34013	66907	42927	84514	69563	28741	76982	48682	03059	75250	21462	88216
90537	19577	93195	83882	95946	51130	80199	02875	23952	37154	36997	12876	77762	49284
04928	21118	41927	15114	57461	38447	90159	94759	70504	68385	18143	82414	17721	21192
01349	32918	90842	52928	77845	13319	44975	29518	56107	79434	68921	15811	47838	77411
80482	89646	25772	70294	46781	80911	06493	89582	29566	35609	58243	11796	55527	04918
33740	60037	11609	59254	27984	30552	65324	27333	61075	44293	10061	60237	86020	23485
35401	20472	21553	93181	22430	96031	09939	07832	64049	39460	99793	02844	10289	31917
53891	71142	38621	72237	81979	64735	49272	90992	98589	21276	85534	89556	05852	15624
79077	04164	01309	36408	20265	94980	10721	87589	53464	60154	45888	15024	97357	74888
87455	58024	90255	55495	92727	25923	60412	48467	28484	99634	21334	44533	28789	58216
92096	17721	77803	30436	42649	89425	17599	71328	64138	17463	80969	63760	64210	10142
49244	44994	87330	65321	35101	58249								

PAST-PAPER HOMEWORK · mini exam

Suggested time: 5 minutes **1 mark** Calculator allowed · show working · hints are below each question if you get stuck

This is the only school-paper item that needs just this lesson's ideas. For more timed practice, do Questions 1, 4(c)(iii) and 7(a) above in 25 minutes.

PP 1 **Blacktown Boys' High School, 2024, HSC Trial Examination, Mathematics Extension 1, Q2** **1 mark**

A standard six-sided die is rolled 18 times.

Let $\hat{p}$ be the proportion of the rolls with an outcome of 3.

Which of the following expressions is the probability that at most 12 of the rolls have an outcome of 3?

A. $P\left(\hat{p} \leq \frac{1}{6}\right)$

B. $P\left(\hat{p} \leq \frac{2}{3}\right)$

C. $P\left(\hat{p} \geq \frac{1}{6}\right)$

D. $P\left(\hat{p} \geq \frac{2}{3}\right)$

Hint: a proportion of the rolls is a relative frequency, like f_r in this lesson.

CHAPTER 13 REVIEW

Chapter review exercise

Mixed practice from the whole chapter, Lessons 13A to 13D.

1 State whether each random variable is numeric or categorical. If it is numeric, say whether it is discrete or continuous.

a The maximum temperature in Sydney on a given day.

b The number of test cricket games in Australia in a given year.

c A die is thrown until it shows a six, and the number of throws required is recorded.

d The state of origin of a Rugby League player.

2 Which of these tables are probability distributions?

a

x	1	2	3	4
$P(X=x)$	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{1}{10}$	$\frac{1}{10}$

b

x	1	2	3	4
$P(X=x)$	0.4	0.1	0.5	0.2

c

x	1	2	3	4
$P(X=x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{1}{6}$	$\frac{1}{8}$

3 Give three reasons why this table is not a valid probability distribution.

x	1	2	3	4
$P(X=x)$	0.2	-0.1	0.5	1.3

4 Copy and complete each table to find the expected value $\mathbf{E}(X)$ of the distribution.

a

x	0	1	2	3	Sum
$p(x)$	0.2	0.3	0.4	0.1	
$xp(x)$					

b

x	-2	-1	0	1	Sum
$p(x)$	0.5	0.1	0.1	0.3	
$xp(x)$					

5 When Jack first visited the Thai Pin Restaurant, he read the menu and assigned each meal a probability indicating how likely he was to order it in the future. This was determined by how much the meal interested him. The fish cost $\$27$ and he rated it $\frac{4}{9}$, the steak cost $\$32$ and he rated it $\frac{2}{9}$, the vegetarian option cost $\$23$ and he rated it $\frac{1}{9}$, the chicken cost $\$25$ and he rated it $\frac{2}{9}$.

a What was Jack's expected cost in buying a meal at the restaurant?

b This is now Jack's favourite restaurant, and he visits it once a week (52 times a year). What is his expected cost over the next year, assuming his interest ratings do not change and the prices remain constant?

6 Copy and complete the probability distribution tables below to calculate $\mathbf{Var}(X)$ using the definition $\mathbf{Var}(X) = \mathbf{E}\left((X-\mu)^2\right)$. Also write down σ.

a

x	1	2	3	4	Sum
$p(x)$	0.4	0.3	0.2	0.1	
$xp(x)$					
$(x-\mu)^2$					—
$(x-\mu)^2p(x)$					

b

x	4	5	6	7	Sum
$p(x)$	0.2	0.6	0.1	0.1	
$xp(x)$					
$(x-\mu)^2$					—
$(x-\mu)^2p(x)$					

7 Now use the alternative formula $\mathbf{Var}(X) = \mathbf{E}(X^2) - \mu^2$ for the variance for the distributions of the previous question. Copy and complete the tables, then calculate the variance.

a

x	1	2	3	4	Sum
$p(x)$	0.4	0.3	0.2	0.1	
$xp(x)$					
$x^2p(x)$					

b

x	4	5	6	7	Sum
$p(x)$	0.2	0.6	0.1	0.1	
$xp(x)$					
$x^2p(x)$					

8 Calculate the mean, variance and standard deviation of each probability distribution.

a

x	0	1	2	3	4
$p(x)$	0.05	0.1	0.8	0	0.05

b

x	0	1	2	3	4
$p(x)$	0.3	0.1	0.2	0.1	0.3

9 Explain briefly the meaning and significance of the expected value of a probability distribution.

10 Explain briefly the meaning and significance of the variance and standard deviation of a probability distribution.

11 For the random variable X, it is known that $\mathbf{E}(X) = 6$ and $\mathbf{Var}(X) = 2$.

a Write down $\mathbf{E}(2X)$, $\mathbf{Var}(2X)$ and σ for the new distribution $2X$.

b Write down $\mathbf{E}(X + 5)$, $\mathbf{Var}(X + 5)$ and σ for the new distribution $X + 5$.

c Write down $\mathbf{E}(3X - 1)$, $\mathbf{Var}(3X - 1)$ and σ for the new distribution $3X - 1$.

12 a The sample space for a certain random variable X is $\{5, 6, 7, 8, 9\}$. The probability distribution is known to be uniform. Write down the table for this probability distribution and find the expected value μ and the standard deviation σ.

b Use random numbers to simulate this experiment. First decide how you will obtain the random numbers, how you will use them, and how many simulations your sample will have. Then draw up a relative frequency table and calculate the sample mean $\bar{x}$ and the sample standard deviation s.